Precipitation and Life Cycle Characteristics of U.S. Mesoscale Convective Systems Simulated in a High-Resolution GFDL AM4

CHANIL PARK,[a] YI MING,[a,b] WENHAO DONG,[c] YI DENG,[d] AND ZHENYU YOU[d]

[a] *Department of Earth and Environmental Sciences, Boston College, Chestnut Hill, Massachusetts*
[b] *Schiller Institute for Integrated Science and Society, Boston College, Chestnut Hill, Massachusetts*
[c] *Department of Earth System Science, Ministry of Education Key Laboratory for Earth System Modeling, Institute for Global Change Studies, Tsinghua University, Beijing, China*
[d] *School of Earth and Atmospheric Sciences, Georgia Institute of Technology, Atlanta, Georgia*

(Manuscript received 3 September 2024, in final form 21 June 2025, accepted 27 July 2025)

ABSTRACT: A high-resolution (~50 km) atmospheric model from GFDL showed promise for simulating observed climatological features of mesoscale convective systems (MCSs) despite nonnegligible biases. This study provides a comprehensive evaluation of precipitation and life cycle characteristics of simulated MCSs in the continental United States. The MCSs are tracked using cloud-top brightness temperature as the sole proxy, enabling independent process-based attribution of MCS biases to their cold cloud systems and precipitation. The MCS precipitation is underestimated in the central United States and overestimated in the eastern United States, with biases being more pronounced in summer than in spring. The pattern of MCS precipitation bias differs from that of total precipitation bias, suggesting that its improvement may not necessarily eliminate the long-standing precipitation deficit in the central United States. Most MCS precipitation is explicitly resolved through large-scale cloud processes. The model well reproduces observed distributions of various MCS diagnostics and their interrelations. However, it produces spurious systems producing very weak and spatially unorganized precipitation (referred to as "dry MCS problem"), which is particularly severe in summer. The linear relation between MCS size and precipitating area is also underestimated, even when the dry MCSs are excluded. While the model accurately captures observed parabolic-shaped MCS size evolution, it misrepresents the fractional precipitating area as a monotonic decrease throughout the life cycle. The MCS precipitation is largely overestimated at initiation, followed by a sharp decrease. This reflects overly efficient precipitation in the convective towers during the upscale growth phase of MCSs, which is hypothesized to be responsible for suppressed stratiform precipitation at maturity.

KEYWORDS: North America; Convective storms/systems; Mesoscale processes; Clouds; General circulation models; Parameterization

1. Introduction

Realistic simulation of hydrological cycles in global climate models (GCMs) is a prerequisite for reliable climate projections and numerical studies of extreme events. The accuracy of modeled precipitation is particularly important, as it is heavily involved in water and energy exchanges within climate systems (Trenberth et al. 2003, 2007; Dai 2006; Schaller et al. 2011). However, sources of model biases in precipitation are often obscured by multiple and nonlinearly intertwined factors (Tapiador et al. 2019). Beyond conventional performance-based evaluation, process- and phenomena-based evaluations are thus gaining popularity because they can provide more direct insights into the physical processes and weather systems associated with simulated precipitation (Leung et al. 2022; Neelin et al. 2023).

Mesoscale convective systems (MCSs) and their related processes could serve as valuable measures for GCM evaluation, as they play pivotal roles not only in local weather extremes (Schumacher and Johnson 2006; Moore et al. 2012; Hu et al. 2021) but also global water and energy budgets (Hartmann et al. 1984; Houze 1989; Stensrud 1996; Stensrud and Anderson 2001; Schumacher et al. 2004; Zhao 2024). The MCSs are contiguous cloud systems consisting of deep convective towers and extensive stratiform clouds. They account for over 50% of year-round tropical precipitation (Nesbitt et al. 2006; Roca et al. 2014) and 30%–70% of warm-season precipitation in the central United States (Fritsch et al. 1986; Feng et al. 2019; Haberlie and Ashley 2019). Their equivalent diameter spans hundreds of kilometers, approaching the grid spacing of conventional GCMs, yet their initiation stems from the aggregation of isolated storms on much smaller scales (Houze 2018). During their life cycle, MCSs exhibit various internal processes on a wide range of scales, including convective updrafts within deep plumes, spreading downdrafts from stratiform clouds, slantwise layered overturning circulations, and formation of gust fronts and cold pools, all of which are closely connected to each other (Smull and Houze 1987; Houze et al. 1989; Houze 1997, Corfidi 2003; Engerer et al. 2008; Moncrieff et al. 2017; see also comprehensive reviews by Houze 2004, 2018; Schumacher and Rasmussen 2020).

The multiscale processes of MCSs are neither resolved nor properly parameterized in conventional GCMs due to their

Supplemental information related to this paper is available at the Journals Online website: https://doi.org/10.1175/JCLI-D-24-0504.s1.

Corresponding author: Chanil Park, parkcha@bc.edu

DOI: 10.1175/JCLI-D-24-0504.1

© 2025 American Meteorological Society. This published article is licensed under the terms of the default AMS reuse license. For information regarding reuse of this content and general copyright information, consult the AMS Copyright Policy (www.ametsoc.org/PUBSReuseLicenses).

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

coarse horizontal resolution and the scale-separation assumption in cumulus parameterizations. Most convection schemes also fail to properly treat the nocturnal or cold-frontal elevated convection above the stable boundary layer, which is common for midlatitude MCSs (Wilson and Roberts 2006; Geerts et al. 2017; Tang et al. 2021), although a few modeling studies tackled this problem (e.g., Park et al. 2024). The consequential misrepresentation of MCSs is considered one of the culprits of inaccurate climate simulation, such as long-standing summertime dry biases over the central United States (Klein et al. 2006; Lin et al. 2017).

Rather than explicitly representing MCSs, Moncrieff et al. (2017) and Moncrieff (2019) parameterized their influence on the resolved-scale circulation using the multiscale coherent structure parameterization. This approach estimates temperature and momentum tendencies associated with the slantwise overturning circulation of MCSs, based on distinct vertical profiles of diabatic heating and vertical momentum transport within MCSs. By incorporating these tendencies alongside those from conventional cumulus parameterization, they enhanced the simulation of the Madden–Julian oscillation and convectively coupled equatorial waves (Yang et al. 2019; Chen et al. 2021), which are generally suppressed when parameterized convection is overly allowed (Kim et al. 2012; Ahn et al. 2019).

Alternatively, cumulus parameterization can be bypassed using the multiscale modeling framework (also known as superparameterization), which replaces the convection scheme with a cloud-resolving (i.e., convection-permitting) model embedded in each column of the host GCM grid (Grabowski and Smolarkiewicz 1999). Despite nonnegligible climatological biases, this method improves MCS-equivalent features and their precipitation (Chen and Kirtman 2018; Lin et al. 2019, 2022; Hsu et al. 2023). However, the multiscale modeling framework does not explicitly represent MCSs when applied to low-resolution (>100 km) GCMs. While it can be applied to higher-resolution GCMs, the computational demands remain a limitation (Lin et al. 2019; Norman et al. 2021; Hsu et al. 2023).

Convection-permitting modeling (horizontal grid spacing < 4 km) allows for explicit representation of convective cloud systems without cumulus parameterization (Tao and Moncrieff 2009; Prein et al. 2015). Recent studies have shown that MCSs and their characteristics (e.g., diurnal cycle) are significantly improved using this modeling approach (Yang et al. 2017; Feng et al. 2018, 2023; Prein et al. 2020; Zhang et al. 2021). However, uncertainties still exist in association with other model physics (e.g., cloud microphysics; Feng et al. 2018; Zhang et al. 2021). The dry biases over the central United States common in contemporary GCMs are also not satisfactorily remedied (Liu et al. 2017; Prein et al. 2020). The current insufficiency of dense observation networks also impedes its process-based evaluation. More importantly, the computational cost is prohibitively high without breakthroughs in computational methods. Therefore, despite its promise, the universal use of convection-permitting modeling for long-term global simulations is unlikely to be feasible in the foreseeable future.

Meanwhile, intermediately high-resolution GCMs can significantly improve the simulation of fine-scale weather systems (Roberts et al. 2018), showing their usefulness for long-term global simulation with a computationally manageable cost. Climate models developed at the Geophysical Fluid Dynamics Laboratory (GFDL) have demonstrated that many aspects of high-impact weather systems, including MCSs, are reasonably well simulated at a horizontal resolution of ~50 km (Zhao et al. 2009, 2010; Murakami et al. 2020; Zhao 2020, 2022; Dong et al. 2021, 2022, 2023; You et al. 2024). A high-resolution atmospheric GFDL GCM is particularly skillful at representing MCSs in both the tropics (Dong et al. 2021, 2022) and midlatitudes (Dong et al. 2023; You et al. 2024).

Focusing on the continental United States, Dong et al. (2023) showed that the GFDL model reliably reproduces the observed size, duration, seasonality, and large-scale environments of warm-season MCSs, although their spatial distribution and diurnal cycle are incorrectly simulated. You et al. (2024) demonstrated that the model captures well the synoptic- and hemispheric-scale circulation patterns associated with springtime MCS initiation. However, the model exhibits positive sea level pressure biases over the North Atlantic subtropical high, potentially contributing to the biased MCS distribution. While basic statistics and large-scale conditions of MCSs were reported, these studies paid limited attention to precipitation which is the most important component of MCSs. The life cycle evolution of MCSs, which is closely related to severe weather, was also not evaluated. For a complete evaluation of MCSs simulated in this model, the present study provides a comprehensive assessment of precipitation and life cycle characteristics of warm-season MCSs over the continental United States.

This paper is organized as follows. Section 2 introduces the observational dataset, model simulation, and MCS tracking procedure. The spatial distribution of MCSs and their precipitation are presented in section 3. The MCS precipitation characteristics and their life cycle evolution are detailed in sections 4 and 5, respectively. Section 6 summarizes and discusses the results.

2. Data and methods

a. *Observational dataset*

The observational data employed for MCS tracking is the 3-hourly infrared brightness temperature T_b retrieved by multisatellites of the Cloud Archive User Service (CLAUS) project with a 1/3° × 1/3° spatial resolution for 1985–2008 (Hodges et al. 2000). Although higher-resolution T_b data are available (e.g., Janowiak et al. 2001), the CLAUS T_b is suitable as its resolution is comparable to that of the model, with a longer overlapping period. The 3-h interval is also sufficient for examining the life cycle characteristics of MCSs (Machado et al. 1998). The MCS precipitation is estimated using the 3-hourly Multisource Weighted-Ensemble Precipitation version 2 (MSWEP V2) with a 0.1° × 0.1° spatial resolution (Beck et al. 2019). All these data are regridded onto

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

the model grids to facilitate a comparison with simulated MCSs.

b. Present-day simulation of GFDL atmospheric GCM

This study uses a high-resolution version of the GFDL Atmospheric Model, version 4 (AM4; Zhao et al. 2018a,b), referred to as C192AM4. The C192AM4 has been utilized for GFDL's participation in the Coupled Model Intercomparison Project phase 6 (CMIP6) High Resolution Model Intercomparison Project (Haarsma et al. 2016). It employs a finite-volume cubed-sphere dynamic core with 192×192 grid boxes per cube face, which corresponds to ~50-km horizontal resolution. As in the default AM4, C192AM4 uses a double-plume convective closure scheme for cumulus parameterization (Zhao et al. 2018b) and a single-moment bulk scheme for cloud microphysics parameterization (Rotstayn 1997; Jakob and Klein 2000). However, a few modifications have been made to this high-resolution version. For example, the model's physics and dynamics time steps were adjusted to be consistent with the increased resolution. The sixth-order divergence damping in the dynamic core was replaced with fourth-order divergence damping to increase the numerical stability. The parameters in the cloud microphysics scheme were also tuned. Readers are referred to Zhao (2020) for details.

We utilize the present-day simulation forced by observed sea surface temperature, sea ice concentration, radiative gases, and aerosol emissions, following the CMIP6 HighResMIP specifications. The simulation spans 1950–2014, but only 1985–2008 is analyzed for consistency with observations. The variables used include the 3-hourly top-of-atmosphere outgoing longwave radiation (OLR), surface precipitation, and its convective (i.e., parameterized) and large-scale (i.e., resolved) components.

c. MCS tracking from observation and C192AM4 output

The MCSs are tracked based on 3-hourly gridded T_b by using the algorithm of Huang et al. (2018). The same algorithm was used by Dong et al. (2023) who evaluated the same model, making their study and ours a coordinated effort to evaluate MCSs simulated in C192AM4. The algorithm first identifies collections of contiguous grid points satisfying $T_b < 221$ K and area coverage $> 30\,000$ km^2 as MCS-candidate cold cloud systems (CCSs) for each time step. These T_b and size thresholds (or similar values) have been used in previous studies to track the Great Plains MCSs (Maddox 1980; Rodgers et al. 1985; Jirak et al. 2003; Cheeks et al. 2020; Dong et al. 2023; Park et al. 2025). The trajectories of individual CCS snapshots are constructed by considering an overlapping ratio of 15% between consecutive time steps. To account for small and fast-moving systems, a Kalman filter is invoked to predict the next location if the overlapping ratio does not meet the threshold. The Kalman filter is successively updated for each track based on the previous traveling speed and direction of CCS. The overlapping approach combined with the Kalman filter better represents the Lagrangian evolution of MCSs. Finally, CCSs lasting at least 6 h (three time steps) are recorded as MCSs. As a postprocessing step, we eliminate systems generated outside the continental United States. Therefore, any hurricanes that could be falsely identified as MCSs are excluded.

Because T_b is not a standard model output, the modeled OLR is converted to T_b using an empirical formula suggested by Ellingson and Ferrado (1983):

$$T_f = T_b(a + bT_b), \tag{1a}$$

$$\mathrm{OLR} = \sigma T_f^4, \tag{1b}$$

where T_f is the flux-equivalent brightness temperature determined by the Stefan–Boltzmann law, $\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4} is the Stefan–Boltzmann constant, and a and b are the fitting parameters set as 1.283 and -1.282×10^{-3} K^{-1}, respectively. The a and b slightly vary for the best fit (Ohring et al. 1984) and may cause systematic biases in the number and T_b of MCSs. The selected values, however, well reproduce the observed T_b climatology over the United States (Dong et al. 2023), ensuring a fair comparison between observed versus simulated MCSs.

Feng et al. (2021a,b) additionally imposed precipitation feature thresholds to exclude weakly precipitating systems associated with purely stratiform clouds. However, we decide to use T_b as the sole proxy for MCS tracking in this study. If both T_b and precipitation are considered, the uncertainties of modeled CCS and precipitation are combined, making it difficult to interpret the source of MCS biases. The precipitation is the final product of numerical modeling and is subject to confounding errors in dynamics and thermodynamics (Tapiador et al. 2019). Recent studies reported that while T_b-alone and T_b-precipitation-combined approaches produce similar MCS climatology in observations with only subtle differences, the latter results in a considerable underestimation of MCSs in the high-resolution Exascale Earth System Model (Leung et al. 2022) and its version 2 (Zhang et al. 2024). This implies that precipitation might not be as tightly involved in CCSs in climate models as in observations. In terms of process-based model evaluation, it would be more instructive to keep the MCS tracking procedure as simple as possible by using T_b as a single proxy and to conduct a post hoc analysis of MCS precipitation (e.g., Hsu et al. 2023).

d. Definition of MCS precipitation

We define MCS precipitation as precipitation occurring within the area of MCSs ($T_b < 221$ K). In midlatitudes, MCSs are often embedded within other precipitating weather systems—such as extratropical cyclones and atmospheric rivers (Moore et al. 2012; Rasp et al. 2016; Tsai et al. 2025)—particularly during spring when baroclinicity and synoptic-scale disturbances are pronounced (Peters and Schumacher 2014; Song et al. 2019; Feng et al. 2019; Park et al. 2025). Acknowledging the inherently multiscale and mixed-type nature of precipitation (Swenson 2023; Konstali et al. 2024; Tsai et al. 2025), we exclude precipitation outside the MCS area to avoid incorporating rainfall exclusively associated with parent systems. This approach differs from that of Hsu et al. (2023) who included precipitation beyond the MCS boundary if the contiguous precipitation objects partially overlap with MCSs.

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

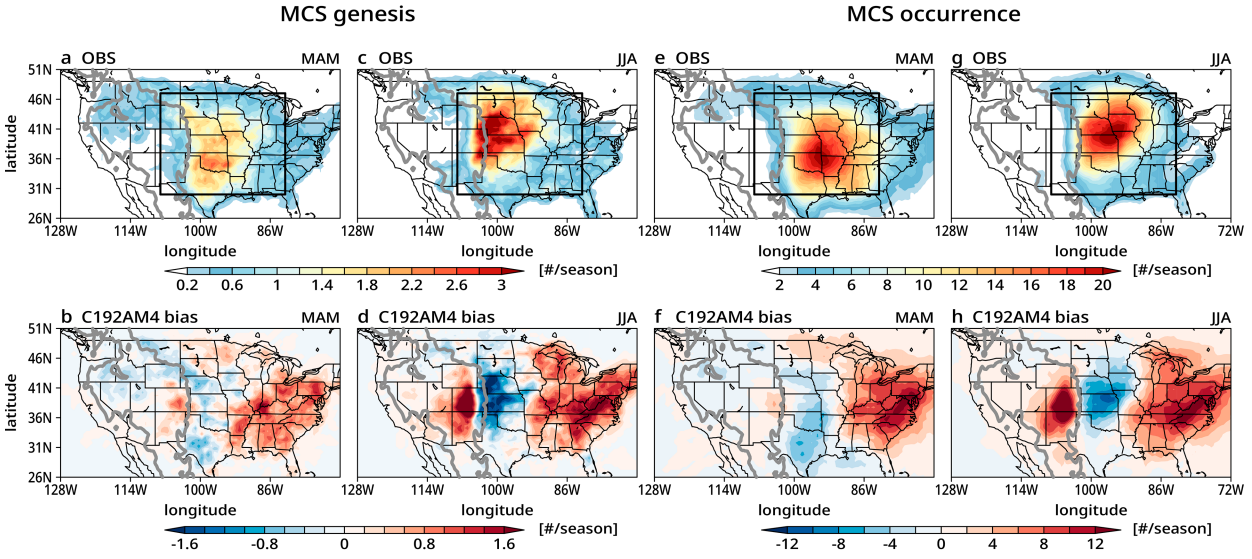

FIG. 1. MCS genesis frequency (number per season) in (a) MAM and (c) JJA, and its model bias in (b) MAM and (d) JJA. (e)–(h) As in (a)–(d), but for MCS occurrence frequency (number per season). The boxed region in (a), (c), (e), and (g) denotes the central United States (252°–277°E, 30°–47°N) for future use. The thick gray contour marks the 1200-m terrain height.

3. Climatological distribution of MCS frequency and precipitation

Climatological distributions of MCSs and their associated precipitation over the continental United States are compared between observations and the C192AM4 simulation. The analysis is confined to the warm season from March to August when MCS activity is pronounced (Schumacher and Johnson 2006). Due to the distinct large-scale environments and thermodynamic characteristics of MCSs across seasons (Peters and Schumacher 2014; Song et al. 2019; Feng et al. 2019), springtime [March–May (MAM)] and summertime [June–August (JJA)] MCSs are examined separately.

Figure 1 depicts the frequencies of MCS genesis and occurrence. The MCS genesis refers to its initial identification, while the occurrence constitutes all 3-hourly records along an MCS track. All grid points meeting the T_b threshold are tallied, accounting for variability in MCS sizes. During spring, MCS genesis is prevalent across the Great Plains (Fig. 1a). The C192AM4 slightly underestimates genesis in this region but substantially overestimates it in the eastern United States (Fig. 1b). In summer, the maximum genesis frequency shifts northward, favoring the Rocky Mountains foothills (Fig. 1c), possibly due to the inland penetration of the North Atlantic subtropical high (Li et al. 2012; Zorzetto and Li 2021). Both the negative bias over the central United States and the positive bias in the east become more pronounced (Fig. 1d). Moreover, the model erroneously produces excessive MCS genesis over the eastern Rockies.

Compared to the MCS genesis frequency, the MCS occurrence frequency exhibits an eastward-smoothed pattern (Figs. 1e,g). The patterns align with previous studies using different observational data and MCS tracking methods (Feng et al. 2019; Haberlie and Ashley 2019; Cheeks et al. 2020). Despite fewer genesis events, the springtime MCS occurrence frequency is comparable to that in summer (cf. Figs. 1a,c,e,g), presumably stemming from the longer lifetimes associated with stronger synoptic-scale forcings (Song et al. 2019; Dong et al. 2023; see also our Fig. 10). The model bias patterns of occurrence frequency resemble those of genesis frequency in both seasons (Figs. 1f,h), suggesting that accurate representation of MCS genesis is paramount for capturing the overall MCS distribution, rather than MCS lifetime or translation speed whose biases are present but subtle (see Figs. 9 and 11 in Dong et al. 2023). Additionally, biases in MCS size may partly contribute to the underestimated MCS occurrence frequency over the Great Plains, as the model tends to underestimate climatological MCS size in the region (Fig. S1 in the online supplemental material). However, the spatial patterns of the size bias are not strongly aligned with those of the frequency bias.

Figures 2a–d depict the spatial distributions of MCS precipitation and the corresponding model biases. In spring, pronounced MCS precipitation is observed across the southern half of the central United States (Fig. 2a), while the summer counterpart exhibits a northward shift (Fig. 2c), mirroring the MCS occurrence frequency patterns. The biases in simulated MCS precipitation also exhibit spatial consistency with those of MCS occurrence frequency (Figs. 2b,d). However, the positive bias over the eastern Rockies in summer is relatively small (Fig. 2d), implying that these spuriously simulated mountainous MCSs are weakly precipitating systems (this will be discussed more later).

Figures 2e–h illustrate the MCS precipitation intensity, defined as accumulated precipitation divided by occurrence frequency. In spring, the highest intensities are observed over the southeastern states (Fig. 2e), which C192AM4 slightly underestimates (Fig. 2f). The model overestimates intensity in the northwestern states where MCSs are rare. In summer, the

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

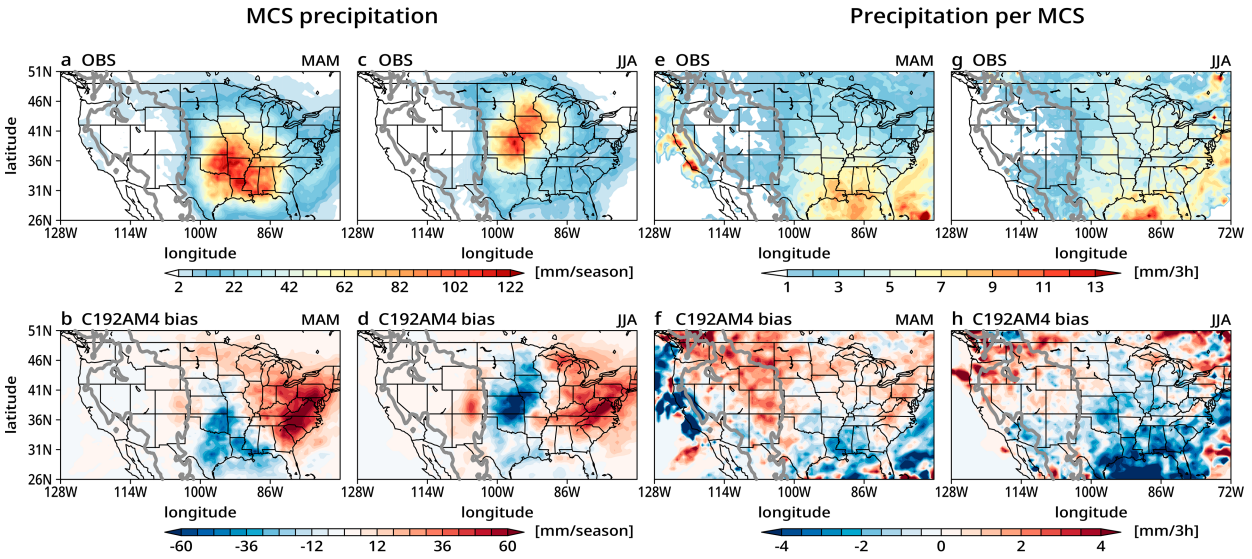

FIG. 2. As in Fig. 1, but for (first column)–(second column) MCS precipitation amount (mm per season) and (third column)–(fourth column) MCS precipitation amount divided by MCS occurrence frequency [mm (3 h)$^{-1}$].

MCS precipitation intensity distribution is more homogeneous than in spring (Fig. 2g), but the model exhibits widespread underestimation with a maximum bias along southern coastal regions (Fig. 2h).

The underestimation of MCS activity over the central United States has been extensively documented in other models with varying configurations (Prein et al. 2020; Feng et al. 2021b; Na et al. 2022; Lin et al. 2022). The overestimation in the eastern United States has also been widely reported, although some models exhibit a nationwide underestimation (Feng et al. 2021b) or a nationwide overestimation (Hsu et al. 2023). However, the overly simulated mountainous MCSs in summer have not been documented before. Such disagreements may also be partly affected by MCS tracking procedures with different proxies and thresholds, requiring systematic intercomparison based on unified criteria (e.g., Prein et al. 2024, Feng et al. 2025).

Figures 3a–d illustrate the fractional MCS precipitation and its model biases. The observed MCSs account for approximately 20%–40% of total precipitation in both seasons (Figs. 3a,c). This fraction is relatively small compared to previous estimations, possibly due to the strict T_b threshold for MCS tracking (Zhang et al. 2024). In spring, the model bias exhibits a similar pattern to that of MCS precipitation amount: negative over the southern half of the central United States and positive in the eastern United States (Figs. 2b and 3b).

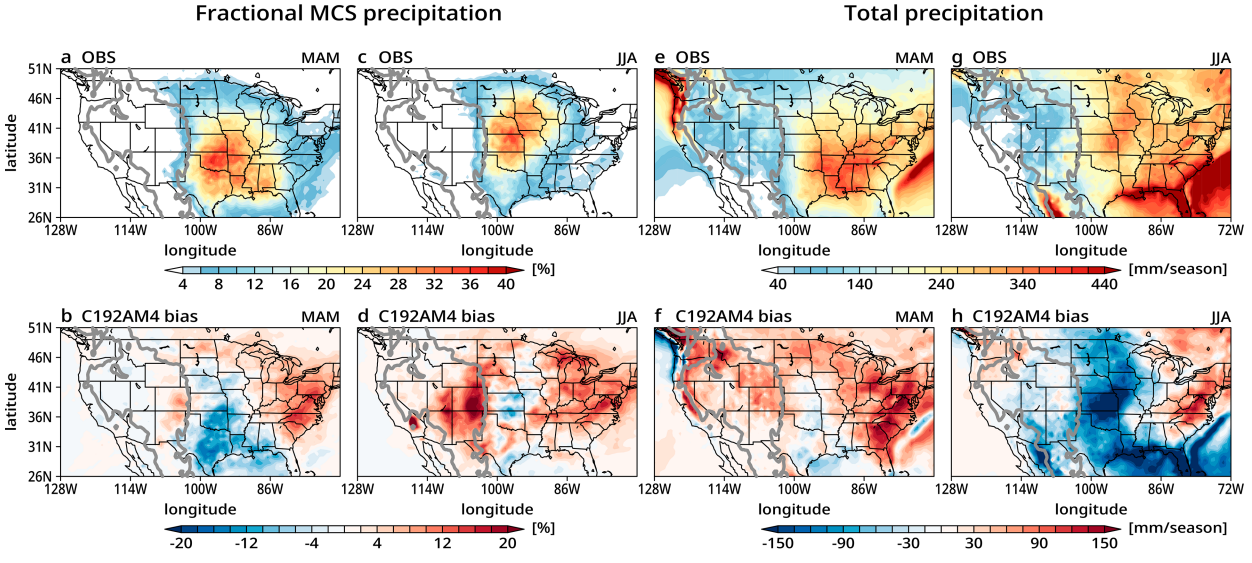

FIG. 3. As in Fig. 1, but for (first column)–(second column) fractional MCS precipitation (%) and (third column)–(fourth column) total precipitation amount not necessarily related to MCSs (mm per season).

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

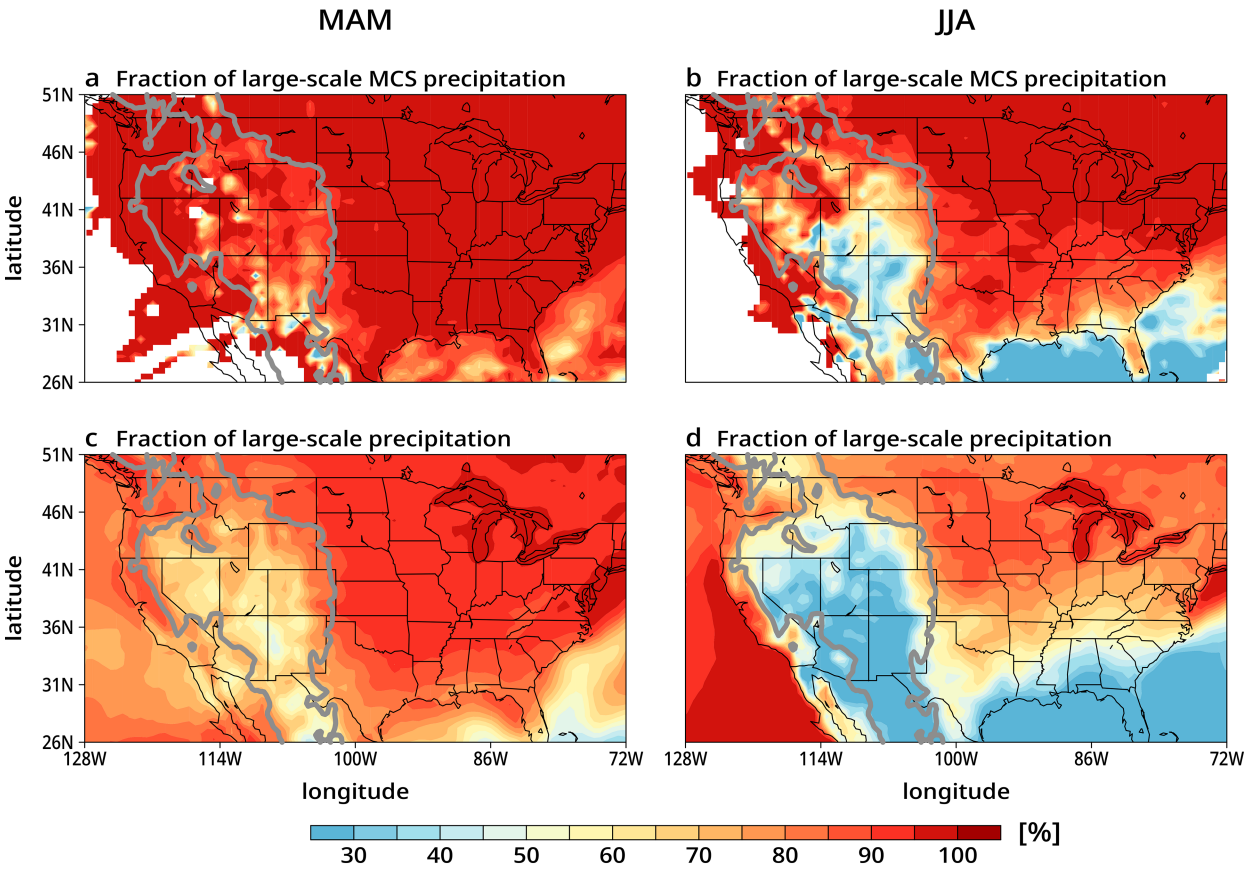

FIG. 4. Fraction (%) of large-scale component of MCS precipitation in C192AM4 in (a) MAM and (b) JJA. (c),(d) As in (a) and (b), but not necessarily related to MCSs. The thick gray contour marks the 1200-m terrain height.

This indicates that the MCS precipitation bias is not explained simply as a part of the total precipitation bias. Indeed, C192AM4 exhibits a relatively weak total precipitation deficit in the southern half of the central United States compared to other regions (Fig. 3f), implying that the negatively biased MCS precipitation in this region is partly compensated by non-MCS precipitation. In contrast, the model has a considerable total precipitation deficit in summer in the central United States (Fig. 3h), contributing to the relatively weak negative bias in the fractional MCS precipitation (Fig. 3d). Instead, the fractional MCS precipitation is positively biased over the Rocky Mountains where total precipitation is slightly negatively biased. This implies that the reasons for the biased MCS and non-MCS precipitation are not identical, and improvement of the former may not ensure the elimination of summertime dry biases in the central United States.

Remarkably, over 90% of simulated MCS precipitation is large-scale (i.e., resolved) precipitation (Figs. 4a,b). This is likely related to the higher spatial resolution of the model than the default AM4 and the significantly increased fractional lateral mixing rate of deep convective plumes compared to previous GFDL models, both of which inhibit parameterized convection (Zhao et al. 2018b). The fraction of large-scale precipitation considerably reduces when including non-MCS precipitation (Figs. 4c,d). This reflects the existence of isolated thunderstorms that are not aggregated into MCSs and thus still required to be parameterized. The result implies the existence of different model processes involved in MCS and non-MCS precipitation, necessitating distinct treatment for their improved simulation. The dominance of resolved precipitation in modeled MCSs further implies that future model development may benefit more from enhancing other physical processes (e.g., cloud microphysics) than from focusing solely on cumulus parameterization.

4. Precipitation characteristics of MCSs

Beyond spatial distributions, MCS precipitation characteristics and their relations to CCS characteristics are compared between observations and C192AM4. To this end, eight MCS diagnostics are defined: 1) MCS size, 2) precipitating area, 3) fractional precipitating area, 4) strongly precipitating area, 5) averaged T_b, 6) minimum T_b, 7) averaged rain rate, and 8) maximum rain rate (see Table 1 for their calculations). To calculate the precipitating area, we aggregate all grid cells within MCSs where precipitation rate exceeds 0.5 mm $(3\ \text{h})^{-1}$. This threshold is more permissive than the 0.5 mm h^{-1} used by Feng et al. (2021b), allowing for a more comprehensive capture of the full spatial extent of MCS precipitation. It is important to note that our definition of precipitating area reflects the

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

TABLE 1. MCS diagnostics employed in this study to statistically evaluate the representation of MCSs in C192AM4. The asterisk denotes the diagnostics automatically computed and saved by the MCS tracking algorithm of Huang et al. (2018). dS_n: pixel area at the nth grid point among N grid points of the instantaneous MCS snapshot. $dA1_n$: pixel area at the nth grid point among M1 grid points satisfying rain rate greater than 0.5 mm (3 h)$^{-1}$ within MCS area. $dA2_n$: pixel area at the nth grid point among M2 grid points satisfying rain rate greater than 12 mm (3 h)$^{-1}$ within MCS area. $T_{b,n}$: brightness temperature at the nth grid point among N grid points of the instantaneous MCS snapshot. P_n: rain rate at the nth grid point among N grid points of the instantaneous MCS snapshot.

MCS diagnostic (unit)	Calculation
*MCS size (km^2)	$\sum_{n=1}^{N} dS_n$
Precipitating area (km^2)	$\sum_{n=1}^{M1} dA1_n$
Fractional precipitating area (dimensionless)	Precipitating area/MCS size
Strongly precipitating area (km^2)	$\sum_{n=1}^{M2} dA2_n$
*Averaged T_b (K)	$\sum_{n=1}^{N} dS_n T_{b,n}$/MCS size
*Minimum T_b (K)	$\min_{1\le n\le N} T_{b,n}$
Averaged rain rate [mm (3 h)$^{-1}$]	$\sum_{n=1}^{N} dS_n P_n$/MCS size
Maximum rain rate [mm (3 h)$^{-1}$]	$\max_{1\le n\le N} P_n$

actual area experiencing precipitation, rather than emphasizing robust, spatially contiguous precipitation features.

For simplicity, the MCS diagnostics are averaged over the MCS lifetime (their lifetime evolutions are examined in section 5). We particularly focus on MCSs generated over the central United States, i.e., the boxed region in Fig. 1. In observations, 1375 and 1720 MCSs are selected for spring and summer, respectively. The C192AM4 overly simulates MCSs in this region with 1738 and 2394 in each season. It will be explained that this discrepancy is primarily due to the overproduction of MCSs bearing feeble precipitation.

a. Probability distribution of MCS diagnostics

Figure 5 compares the probability density functions (PDFs) of MCS size, precipitating area, fractional precipitating area, and strongly precipitating area between observed and simulated MCSs. Note that MCSs are tracked solely using T_b, and thus, MCS size refers to the areal extent of CCS (i.e., precipitating plus nonprecipitating anvil regions). The C192AM4 reasonably well represents the observed MCS size distribution with a maximum probability in the range of 6×10^4–9×10^4 km^2 in both seasons, although small-size MCSs are slightly overly simulated (Figs. 5a,b). The model exhibits a noticeable underestimation of the precipitating area, especially in summer (Figs. 5c,d). It should be noted that these biases are not systematic. In terms of number distributions (not PDFs), C192AM4 overproduces MCSs with small sizes or precipitating areas but produces a comparable number of MCSs with moderate-to-large sizes and precipitating areas compared to observations.

The modeled MCSs exhibit smaller fractional precipitating areas than observed MCSs in both seasons, with the bias being more pronounced in summer (Figs. 5e,f). The PDFs of modeled MCSs have a secondary peak at fractional precipitating area = 0–0.1. In contrast to the whole precipitating area, the model relatively well captures the strongly precipitating area distribution in spring (Fig. 5g). However, the model still severely overproduces MCSs with strongly precipitating area less than 0.5×10^4 km^2 in summer (Fig. 5h).

Figures 6a–d compare the PDFs of averaged T_b and minimum T_b between observed and modeled MCSs. While the averaged T_b is related to overall cloud-top heights of MCSs, the minimum T_b is more tied to the height of local convective towers within MCSs. Both are significantly overestimated in C192AM4, with the biases more pronounced in summer. This implies that simulated MCSs have lower cloud-top heights with weaker convective penetration than observed MCSs. Figures 6e–h compare the PDFs of averaged and maximum rain rates between observed and simulated MCSs. While C192AM4 reasonably well captures the distribution of the averaged rain rate in spring (Fig. 6e), it is largely underestimated in summer, particularly due to an overproduction of MCSs with averaged rain rates less than 2 mm (3 h)$^{-1}$ (Fig. 6f). The distribution of maximum rain rate is also reasonably well captured by C192AM4 in spring, but the overproduction of MCSs with maximum rain rate less than 4 mm (3 h)$^{-1}$ is present, particularly in summer (Figs. 6g,h).

The PDF analyses reveal that C192AM4 overproduces MCSs with very weak and spatially unorganized precipitation, which are rare in observations. We refer to this model defect as the "dry MCS problem." It is important to emphasize that this issue would not be identified if precipitation were combined with T_b as joint proxies for MCS tracking (e.g., Feng et al. 2021a,b). To examine the influence of dry MCSs on the statistical distributions of MCS diagnostics, we repeat the PDF analyses after excluding such dry MCSs. The dry MCSs are defined as those with precipitating areas $<3 \times 10^4$ km^2, averaged rain rates <2 mm (3 h)$^{-1}$, and maximum rain rates <4 mm (3 h)$^{-1}$, with all being lifetime means. These thresholds correspond to the leftmost bins of the PDF analyses, but their slight changes do not affect the results. It is found that 15.0% and 26.9% of simulated MCSs in spring and summer, respectively, are counted as dry MCSs, while only 5.4% and 1.1% are dry MCSs in observations for each season. Without the dry MCSs, the total number of observed and simulated MCSs becomes more comparable (1301 and 1478 MCSs in spring and 1701 and 1751 MCSs in summer for observations and model, respectively). The distributions of modeled MCS diagnostics also become closer to the observed distributions (Figs. S2 and S3).

b. Statistical relations between MCS diagnostics

In this subsection, we examine if the model reproduces the observed relationships between different MCS diagnostics. We particularly focus on the CCS–precipitation coupling

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

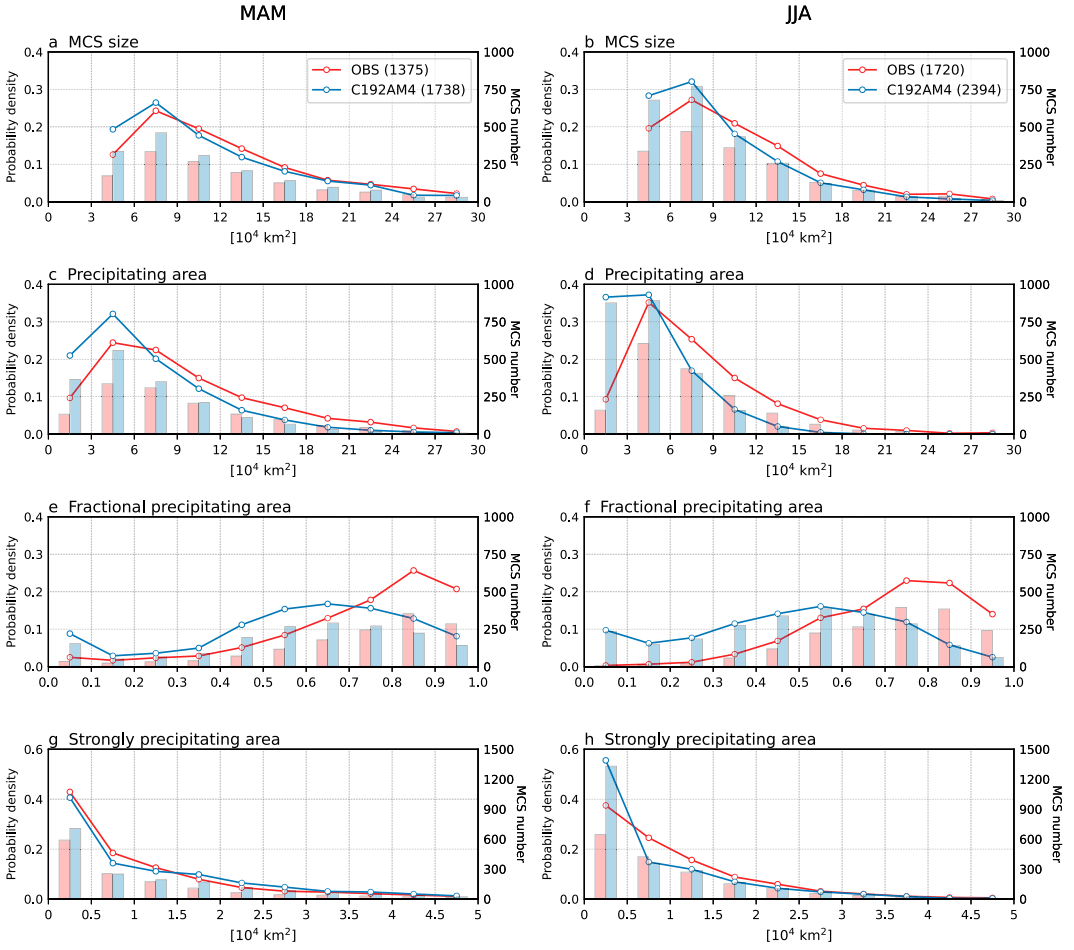

FIG. 5. PDFs (lines; left y axis) and number distribution (bars; right y axis) of MCSs as a function of (a),(b) MCS size, (c),(d) precipitating area, (e),(f) fractional precipitating area, and (g),(h) strongly precipitating area for observation and C192AM4 in (left) MAM and (right) JJA. All quantities are lifetime-mean values. The MCSs generated in the boxed region in Fig. 1 are considered. Also printed in the legend is the number of the considered MCSs.

within MCSs. Figure 7 shows the joint PDFs of MCSs on the plane of MCS size and precipitating area for observed and modeled MCSs. In observations, the precipitating area is linearly proportional to MCS size, with best-fit slopes and correlation coefficients of 0.81 and 0.93 in spring (Fig. 7a) and 0.75 and 0.91 in summer (Fig. 7b). This relationship is weaker in C192AM4 with gentler slopes (Figs. 7c,d). This indicates that spatially large MCSs in the model do not accompany widespread surface precipitation to the same extent as observed. Notably, modeled MCSs exhibit two distinct regimes: one along the linear best fits and another characterized by very small precipitating areas, implying that dry MCSs are not simply the end of MCS precipitation distribution but spurious systems that should be eliminated through model improvements. However, even when the dry MCSs are excluded, the underestimated relation persists with best-fit slopes deviating further from observations (cf. blue and green dashed lines). This means that the weaker size–precipitating area relation in the model is not the result of dry MCS problem but ubiquitous for all simulated MCSs.

Figure 8 represents the joint PDFs of MCSs on the plane of averaged T_b and averaged rain rate. The observed MCSs do not exhibit a significant correlation between these two diagnostics in both seasons (Figs. 8a,b). The lack of correlation perhaps reflects the existence of stratiform clouds that result from the weakening of old convective cells into a layered structure but still accompany substantial precipitation over a wide area (Houze 1997). In contrast, they are negatively correlated in the model, with best-fit slopes of -0.43 mm (3 h)$^{-1}$ K^{-1} and correlation coefficients of -0.35 in spring (Fig. 8c) and -0.37 mm (3 h)$^{-1}$ K^{-1} and -0.28 in summer (Fig. 8d). The maximum joint probability of modeled MCSs occurs for very weak averaged rain rates, i.e., the dry MCS problem. Eliminating dry MCSs slightly weakens but does not eliminate the negative correlation between averaged T_b and averaged rain rate (cf. blue and green dashed fitted lines).

Figure 9 depicts the joint PDFs of MCSs on the plane of minimum T_b and maximum rain rate. In contrast to the lack of a significant relationship between averaged T_b and averaged rain rate shown in Figs. 8a and 8b, the minimum T_b and maximum rain rate are negatively correlated with best-fit slopes of -0.91 mm (3 h)$^{-1}$ K^{-1} and correlation coefficients of -0.44 in spring (Fig. 9a), and -0.7 mm (3 h)$^{-1}$ K^{-1} and

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

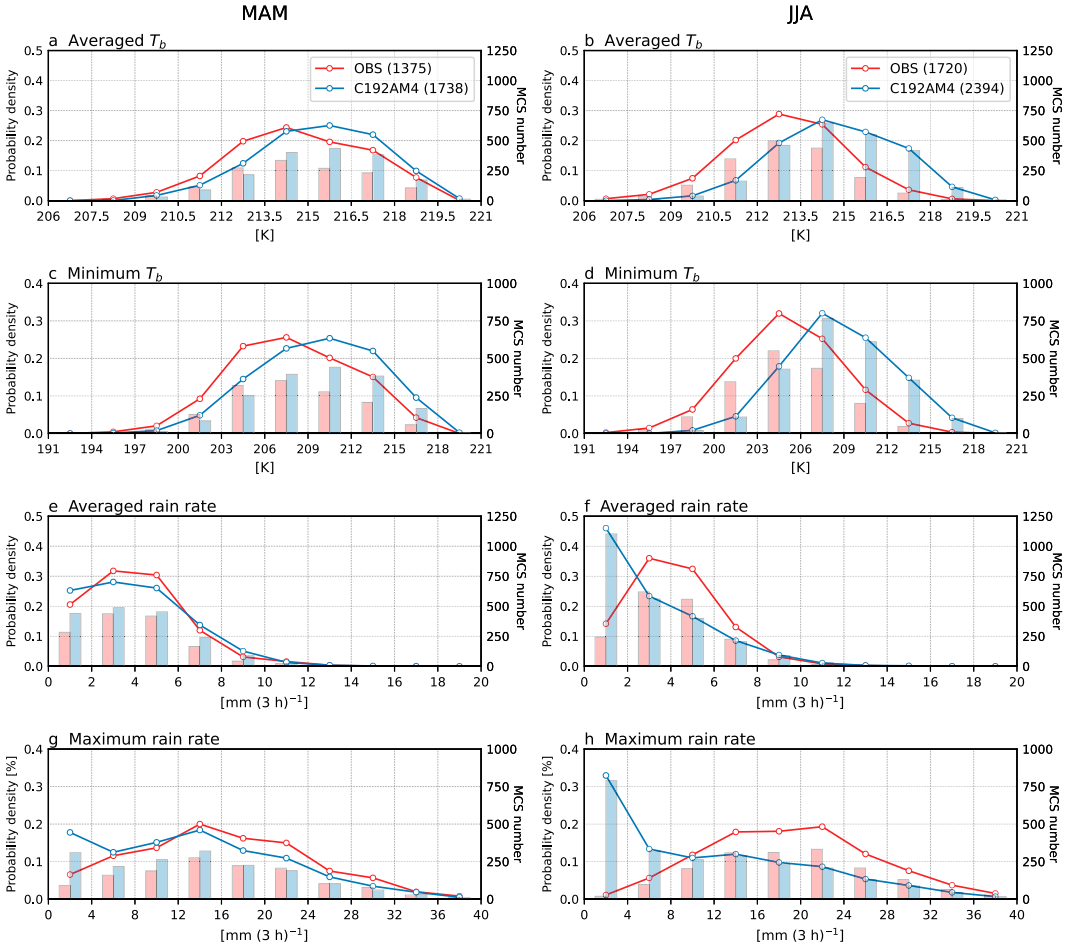

FIG. 6. As in Fig. 5, but for (a),(b) averaged T_b, (c),(d) minimum T_b, (e),(f) averaged rain rate, and (g),(h) maximum rain rate.

−0.31 in summer (Fig. 9b). This indicates that the strength of local convective penetration partly determines the local precipitation maximum. The model simulation captures this relationship but with larger negative correlation coefficients and steeper best-fit slopes (Figs. 9c,d). The linear relation is still observed with best-fit slopes and correlation coefficients closer to observations when dry MCSs are excluded (cf. blue and green dashed fitted lines).

It is further found that averaged and minimum T_b are very tightly correlated with correlation coefficients larger than 0.9 in both seasons (Figs. S4a,b), indicating that the strength of local convective penetration determines the overall height of CCSs. The averaged and maximum rain rates are also strongly correlated with correlation coefficients of 0.82 in spring and 0.77 in summer (Figs. S5a,b). The model skillfully captures these relations with best-fit slopes and correlation coefficients close to observations regardless of the inclusion of dry MCSs (Figs. S4c,d and S5c,d).

5. Life cycle characteristics of MCSs

The MCSs exhibit notable variations in their characteristics during their life cycle (Loehrer and Johnson 1995; Machado et al. 1998), which is closely associated with weather hazards (Johns and Doswell 1992; Gallus et al. 2008; Doswell et al. 1996). In this section, we evaluate the life cycle evolution of modeled MCSs using the diagnostics listed in Table 1. Figure 10 shows that the MCS lifetime is widely distributed, with short-lived MCSs being more popular, particularly in summer. The model well reproduces the observed MCS lifetime distributions. The shortest lifetime of tracked MCSs is 6 h due to the lifetime threshold in MCS tracking, providing at least three time steps (based on 3-hourly records) to examine MCS evolution. Given the varying lifetimes across MCSs, we express the life cycle evolution as a function of normalized MCS time frames. It is found that dry MCSs are predominantly short lived (Fig. S6). Their precipitation-related diagnostics have negligible magnitudes throughout the MCS lifetime (not shown). The MCS size, averaged T_b, and minimum T_b evolutions also exhibit very small life cycle amplitudes (Fig. S7). Therefore, for a fair comparison, we exclude dry MCSs from the following analyses.

a. Life cycle evolutions of MCS diagnostics

Figure 11 illustrates the life cycle evolutions of MCS size, precipitating area, fractional precipitating area, and strongly

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

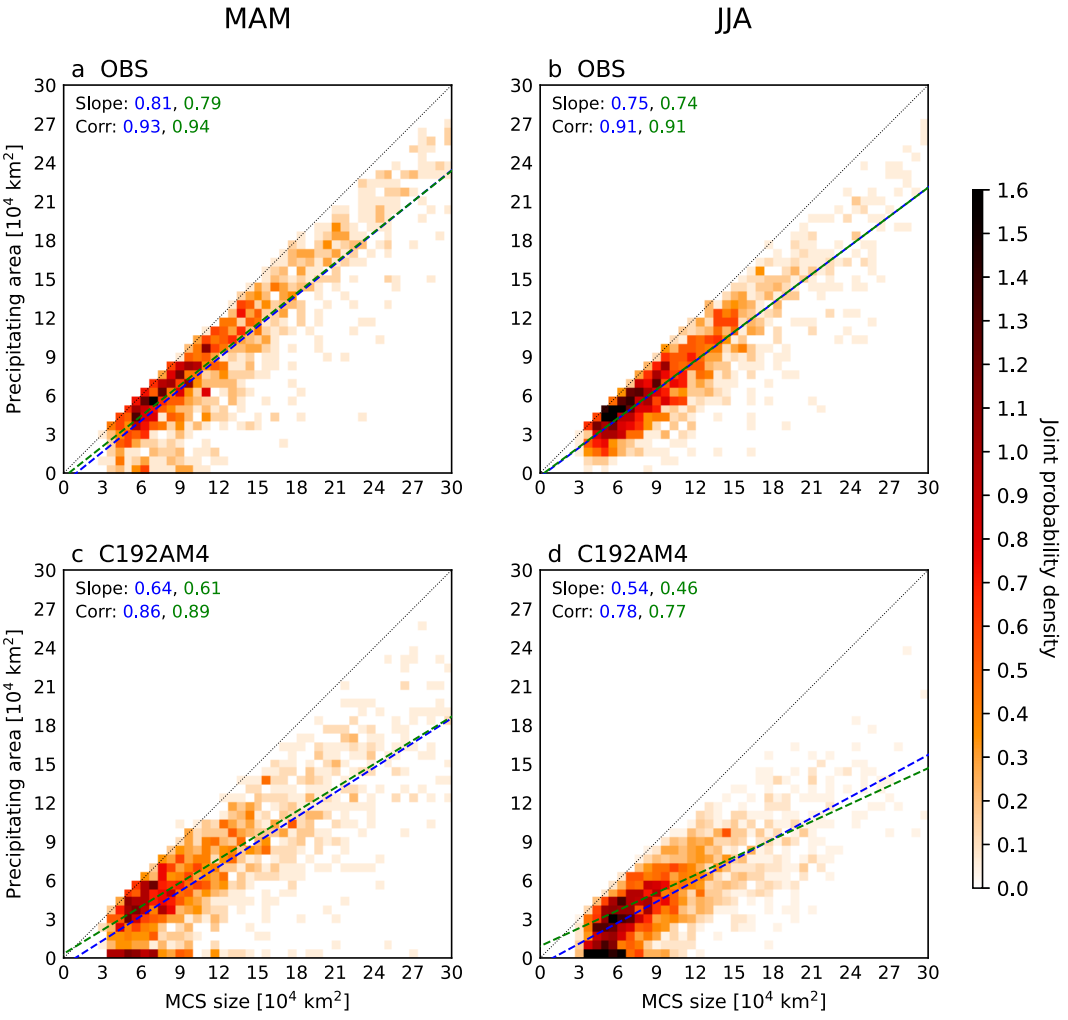

FIG. 7. Joint PDFs of observed MCSs in the coordinate where the *x* axis is lifetime-mean MCS size and the *y* axis is lifetime-mean precipitating area in (a) MAM and (b) JJA. (c),(d) As in (a) and (b), but for MCSs in C192AM4. The blue and green dashed lines, respectively, denote the linear best fits with and without dry MCSs. Also printed in each panel are the regression and correlation coefficients for the two linear fits. The MCSs generated in the boxed region in Fig. 1 are considered.

precipitating area. The observed MCS size exhibits a parabolic evolution with maximum size at the midpoint of the lifetime (Figs. 11a,b), consistent with previous observational studies (Machado et al. 1998; Cheeks et al. 2020). Springtime MCSs attain larger sizes in spring than their summer counterparts, possibly due to stronger synoptic-scale forcing (Peters and Schumacher 2014; Feng et al. 2019). The C192AM4 successfully represents the observed MCS size evolution, closely matching observations in the timing of upscale growth, maturity, and decay. This implies that the model is skillful at representing the development of stratiform and anvil clouds following the initial organization of deep convection. Hereafter, we refer to the timings of increasing, peak, and decreasing MCS size as upscale growth, mature, and decaying phases, respectively.

The observed precipitating area follows a similar parabolic curve as MCS size (Figs. 11c,d). While the modeled initial precipitating area matches observations ($\sim 6 \times 10^4$ km^2), it is subsequently underestimated primarily due to a smaller increasing rate during the upscale growth phase. Consequently, the simulated fractional precipitating area monotonically decreases throughout the life cycle from 0.8 to 0.5 in spring and from 0.8 to 0.4 in summer, while the observed fractional precipitating area almost keeps the initial value with very slow decreasing rates (Figs. 11e,f). This suggests that the weaker MCS size–precipitating area relationship in the model (see Fig. 7) mostly stems from the underestimated growth rate of precipitating area during the upscale growth phase of MCSs.

The modeled strongly precipitating area shows substantial discrepancies compared to observations (Figs. 11g,h). While observed MCSs exhibit a parabolic evolution that is in phase with MCS size, this feature is absent in the model. In spring, the model overestimates the strongly precipitating area at initiation, reaching its peak earlier than observed, followed by a sharp decline (Fig. 11g). Both observations and C192AM4

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

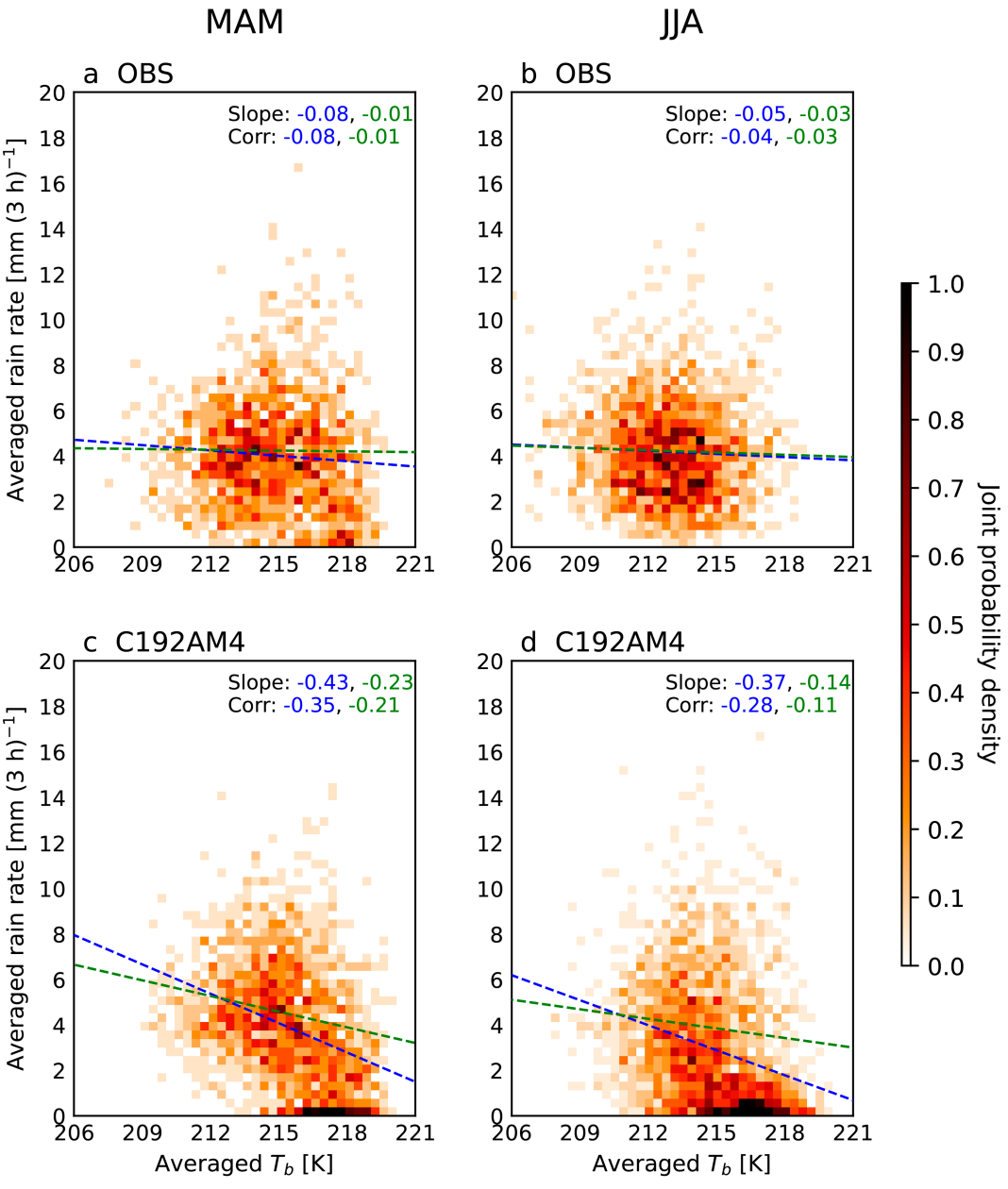

FIG. 8. As in Fig. 7, but on the plane of lifetime-mean averaged T_b (x axis) and lifetime-mean averaged rain rate (y axis).

show considerable case-to-case variability, as indicated by the wide interquartile ranges. In summer, the modeled strongly precipitating area starts at a level comparable to observations (Fig. 11h), but it is still overestimated and again peaks earlier, followed by a rapid decrease.

Figure 12 depicts the life cycle evolutions of averaged T_b, averaged rain rate, minimum T_b, and maximum rain rate. Unlike MCS size, the observed averaged T_b and minimum T_b weakly decrease during the upscale growth phase and steeply increase during the decaying phase of MCSs (Figs. 12a–d), consistent with the prior climatological study by Machado et al. (1998). The C192AM4 captures these asymmetric evolutions in both seasons but overestimates both averaged T_b and minimum T_b to a greater extent in summer. While the model systematically overestimates averaged T_b and minimum T_b throughout the life cycle in summer (Figs. 12b,d), their initial values are accurately captured and subsequently overestimated in spring (Figs. 12a,c).

A stark misrepresentation is found in the evolution of the averaged rain rate (Figs. 12e,f). While the observed averaged rain rate keeps its initial values with a slight decrease in the last of the life cycle in both seasons, the modeled averaged rain rate is largely overestimated initially and sharply decreases afterward. The maximum rain rate of observed MCSs increases and decreases during upscale growth and decaying phases, respectively, in phase with MCS size evolution (Figs. 12g,h). The

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

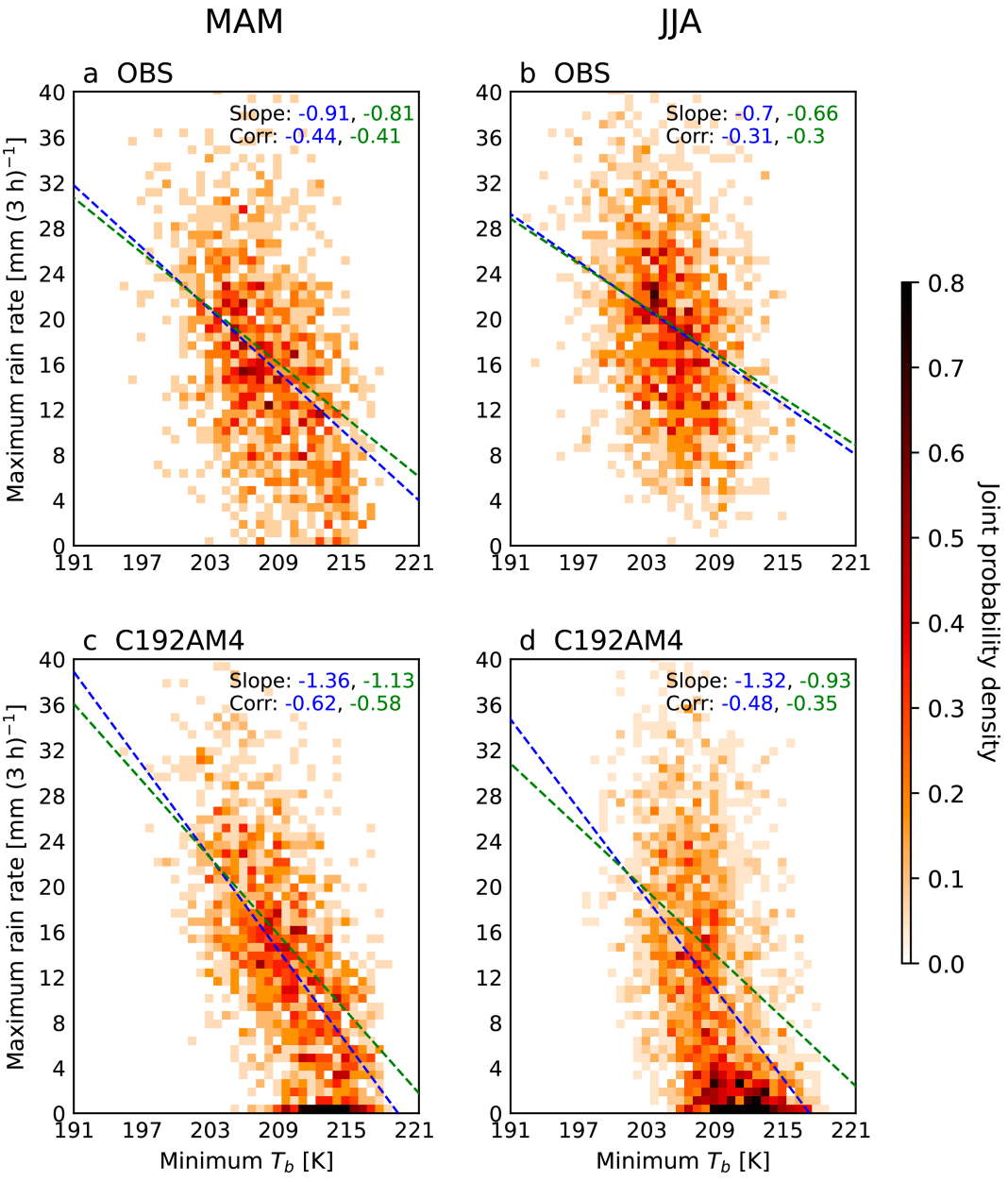

FIG. 9. As in Fig. 7, but on the plane of lifetime-mean minimum T_b (x axis) and lifetime-mean maximum rain rate (y axis).

model reasonably captures the initial maximum rain rate but misses its increasing feature, leading to significant underestimation during the remaining life cycle.

The initially overestimated averaged rain rate reflects excessive precipitation in the convective towers of MCSs during the upscale growth phase. Given the accurate simulation of the initial maximum rain rate (Figs. 12g,h) and the overestimation of the initial strongly precipitating area (Figs. 11g,h), this bias likely arises from the oversize of convective plumes rather than their local precipitation maxima. Such overly efficient precipitation can rapidly deplete ambient water vapor, thereby reducing the number of hydrometeors available for detrainment into stratiform clouds during mature and decaying phases of MCSs (Houze 1997; Elsaesser et al. 2022). The resulting atmospheric drying may also enhance the reevaporation of raindrops within stratiform regions, whose efficiency is parameterized in the model to be limited by a prescribed threshold of relative humidity (Rotstayn 1997). These model behaviors may also explain the monotonic decreases in fractional precipitating area (Figs. 11e,f) and averaged rain rate (Figs. 12e,f).

b. *Sensitivity to MCS lifetime*

The dynamic and thermodynamic characteristics of long-lived MCSs differ from those of short-lived MCSs (Machado et al. 1998; Coniglio et al. 2010), and numerical models do not

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

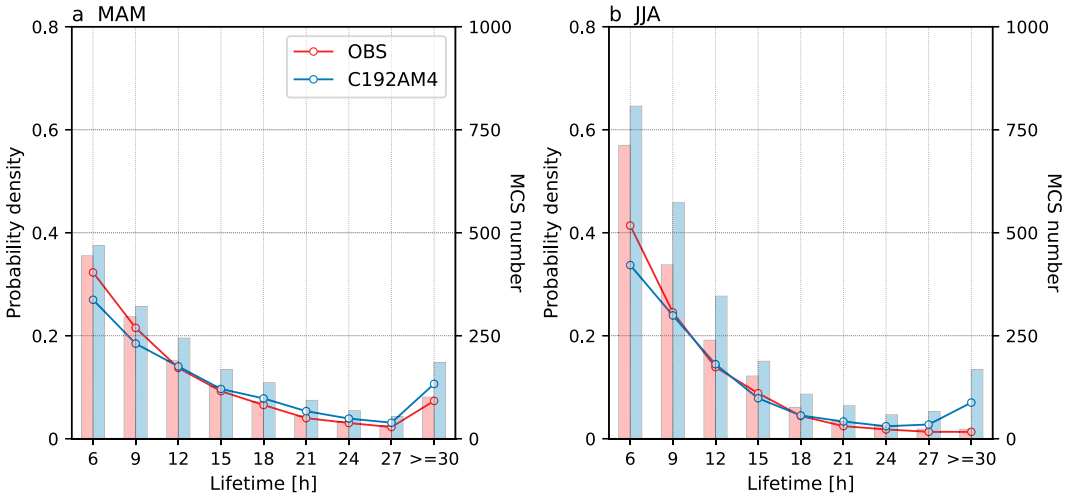

FIG. 10. PDFs (lines; left y axis) and number distributions (bars; right y axis) of MCS lifetime for observation and C192AM4 in (a) MAM and (b) JJA. The MCSs generated in the boxed region in Fig. 1 are considered.

necessarily represent their life cycle evolutions similarly (e.g., Prein et al. 2020; Zhang et al. 2021). The MCS life cycle evolutions are further compared between observations and C192AM4 by separately considering short-lived (lifetime $\leq$ 9 h) and long-lived (lifetime $\geq$ 12 h) MCSs. Here, MCSs in spring and summer are not distinguished for simplicity, as their seasonal differences fall within the range of differences depending on MCS lifetime. We primarily discuss MCS size, averaged rain rate, strongly precipitating area, and maximum rain rate (Fig. 13), with the other diagnostics present as supplemental material (Fig. S8).

In observations, long-lived MCSs attain a larger size than short-lived MCSs (Figs. 13a,b), consistent with the well-documented size–lifetime relation (Machado et al. 1998; Cheeks et al. 2020). The C192AM4 successfully captures this feature, although it slightly underestimates the long-lived MCS size at maturity (Fig. 13b). For both short-lived and long-lived MCSs, the model underestimates the increasing precipitating area during the upscale growth phase (Figs. S8a,b), resulting in the monotonically decreasing fractional precipitating area (Figs. S8c,d). The averaged rain rate of modeled MCSs is also largely overestimated at initiation but decreases throughout the life cycle for both MCS types (Figs. 13c,d), implying too efficient initial precipitation followed by suppressed stratiform precipitation in mature and decaying phases. The evolutions of these MCS diagnostics in observations and the model are quite consistent with the previous subsection where MCSs are not separated by their lifetimes.

On the contrary, the model differently simulates the life cycle evolutions of strongly precipitating area and maximum rain rate between short-lived versus long-lived MCSs (Figs. 13e–h). In observations, the evolutions of both diagnostics are in phase with the MCS size evolution with greater peaks in long-lived MCSs than in short-lived counterparts. The greater peaks may be associated with typical environmental conditions of long-lived MCSs, which allow the continuous redevelopment of convective cells, e.g., large-scale forced ascent, sufficient moisture supply, a well-defined frontal zone, and proper low-level wind shear (Crook and Moncrieff 1988; Rotunno et al. 1988; Weisman et al. 1988; Raymond and Jiang 1990; Coniglio et al. 2010; Yang et al. 2017; Park et al. 2025). In the model, the increasing features of strongly precipitating area and maximum rain rate are captured only for long-lived MCSs with their peaks appearing earlier than observations. For simulated short-lived MCSs, these two diagnostics exhibit monotonic decreases throughout the life cycle. While further analysis is required, our interpretation is that favorable environmental conditions of long-lived MCSs partly support the development of cumulus convection during the upscale growth phase (the decreasing minimum T_b during the upscale growth phase is also captured only for long-lived MCSs; see Figs. S8g,h) but seem still insufficient to overcome too efficient early stage precipitation.

6. Summary and discussion

The present study evaluates the precipitation and life cycle characteristics of warm-season (March–August) MCSs over the continental United States simulated in a high-resolution GFDL AM4, i.e., C192AM4. We track MCSs using cloud-top brightness temperature T_b as a sole proxy without consideration of precipitation, facilitating an independent evaluation of CCSs and precipitation related to MCSs. Building upon previous studies that evaluated other aspects of MCSs in this model (Dong et al. 2023; You et al. 2024), these studies will serve as useful references for future model development. Our key findings are summarized below.

Consistent with many other climate models, C192AM4 underestimates MCS frequency in the central United States and overestimates it in the eastern United States, with the former being more severe in summer. Consequently, there is an MCS precipitation deficit in the central United States and a surplus in the eastern United States. The MCS precipitation bias is not explained simply as part of the total precipitation bias, implying that reducing the MCS precipitation bias may

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

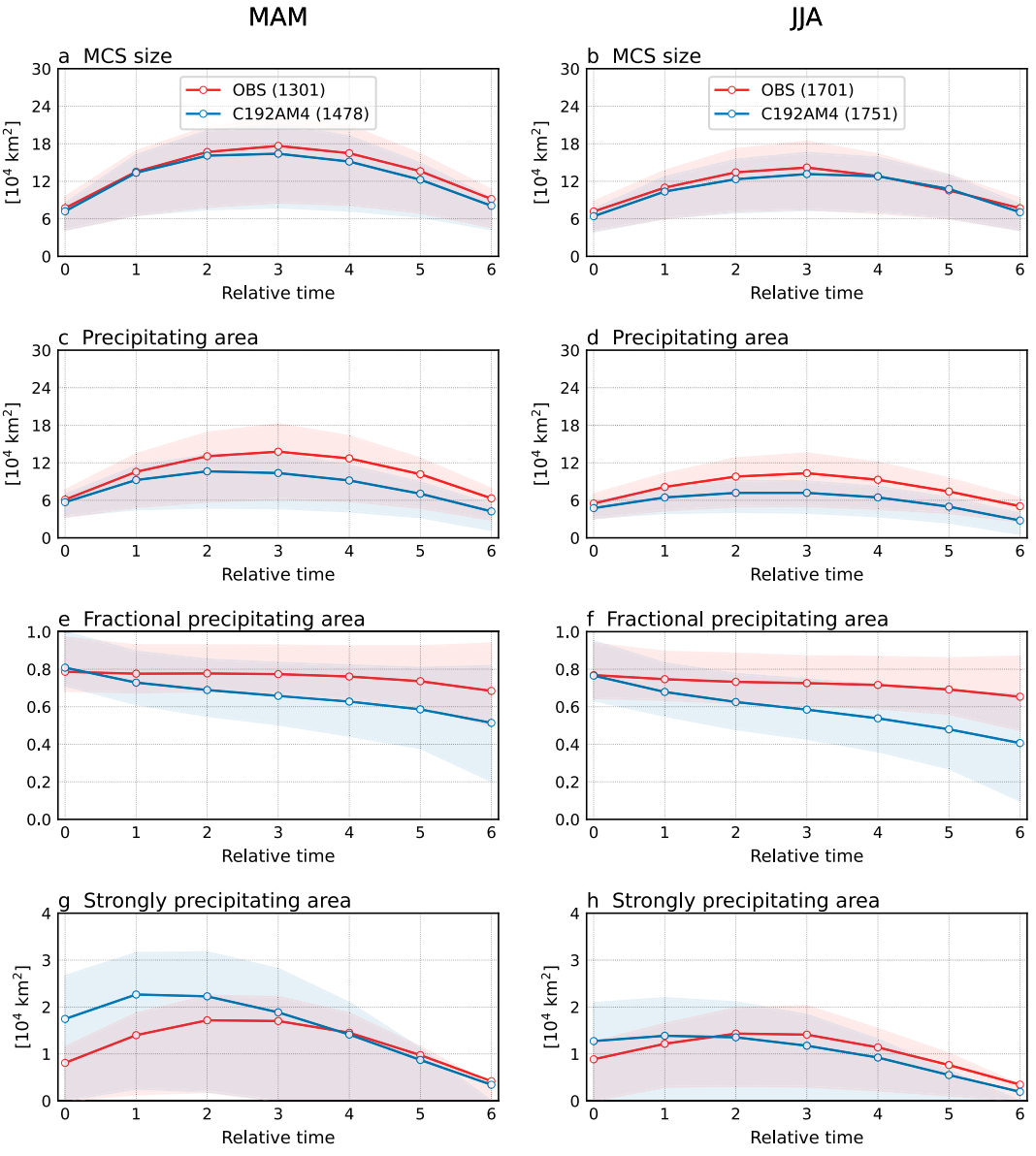

FIG. 11. Composite life cycle evolutions of (a),(b) MCS size, (c),(d) precipitating area, (e),(f) fractional precipitating area, and (g),(h) strongly precipitating area in (left) MAM and (right) JJA. The dry MCSs are excluded. The x axis is the normalized MCS time, and the colored shadings denote the 25th–75th percentile ranges. The MCSs generated in the boxed region in Fig. 1 are considered. Also printed in the legend is the number of the considered MCSs.

not ensure the improvement of the non-MCS precipitation bias. We also find that C192AM4 excessively simulates MCSs over the eastern Rockies in summer. Over 90% of MCS precipitation is explicitly resolved, possibly because the parameterized connection is largely inhibited in this model (Zhao et al. 2018b).

We further detail the precipitation characteristics of MCSs in the model based on MCS diagnostics listed in Table 1. The C192AM4 skillfully represents the observed MCS size distribution. However, the linear correlation between MCS size and precipitating area is weaker than that in observations, resulting in a significant underestimation of fractional precipitating area. Compared to observations, the modeled MCSs are less penetrative (i.e., higher T_b), especially in summer. Importantly, the model overly simulates MCSs producing very weak and spatially unorganized precipitation, termed the "dry MCS problem." Approximately 15.0% and 26.9% of modeled MCSs over the central United States are dry MCSs in spring and summer, respectively, which are much larger fractions than observations (5.4% and 1.1%). Excluding dry MCSs, the model represents the distributions of MCS diagnostics and their interrelations closer to observations, except for the MCS size–precipitating area relation which is still weaker than observations. The cause of the dry MCS problem

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

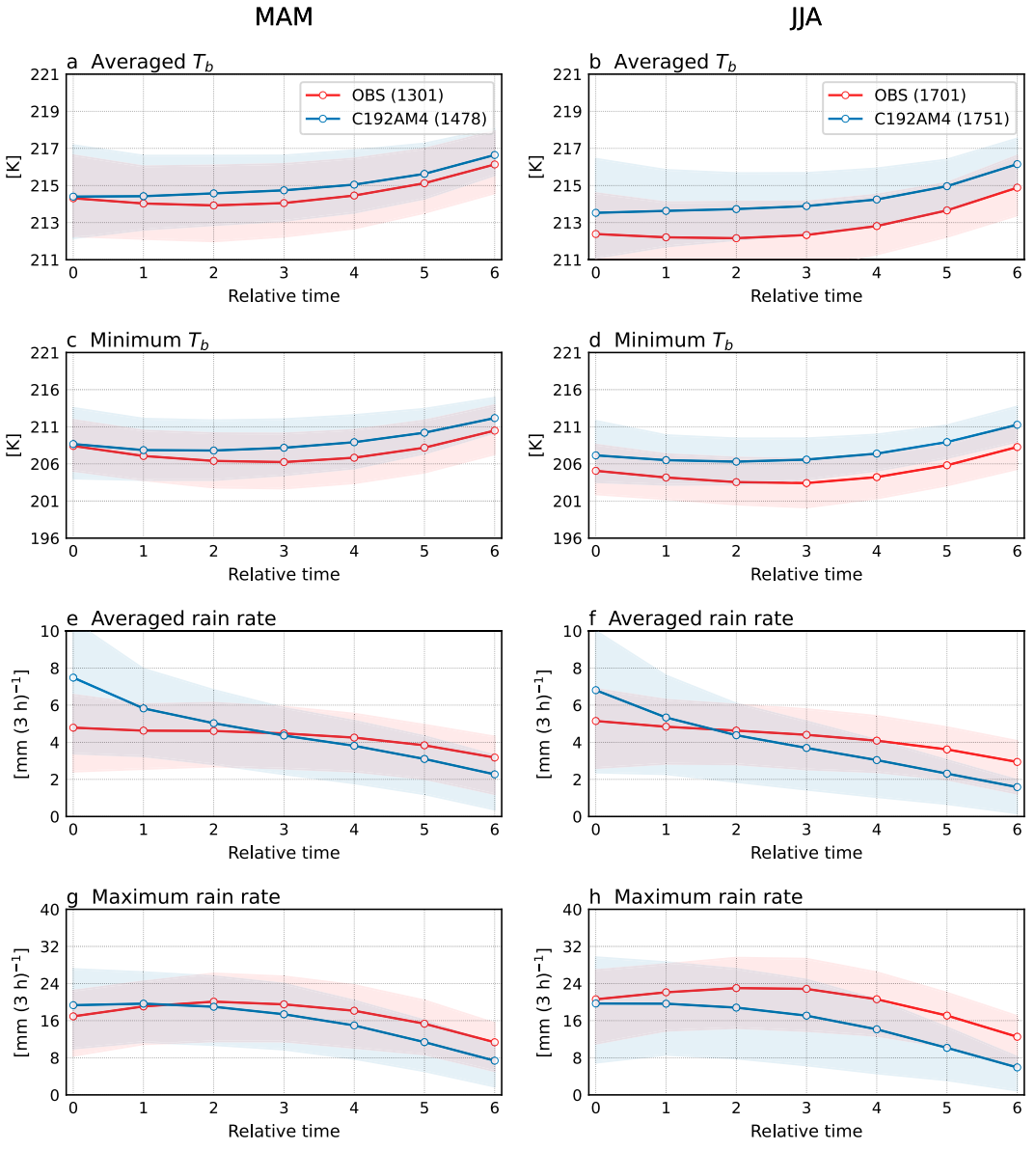

FIG. 12. As in Fig. 11, but for (a),(b) averaged T_b, (c),(d) minimum T_b, (e),(f) averaged rain rate, and (g),(h) maximum rain rate.

is currently unclear and will be investigated in a future study by conducting experiments with perturbed or different model physics (e.g., Guo et al. 2021).

The model successfully reproduces the parabolic-shaped evolutions of MCS size and precipitating area. However, it underestimates the increasing tendency of precipitating area during the upscale growth phase. Consequently, the fractional precipitating area monotonically decreases in the model, contrasting with that in observations, which maintains the initial value throughout the life cycle. The averaged rain rate is substantially overestimated at MCS initiation and sharply decreases afterward, while the observed counterpart remains relatively stable. We hypothesize that this misrepresented averaged rain-rate evolution is related to overly efficient precipitation in the convective cores of MCSs during their upscale growth phase, which leads to too fast depletion of ambient water vapor. The model fails to capture the increasing tendency of strongly precipitating area and maximum rain rate during the upscale growth phase for short-lived MCSs. For long-lived MCSs, however, these tendencies are partially simulated. Given that long-lived MCSs are generally supported by stronger synoptic forcings for low-level moisture transport, this difference aligns with our hypothesized microphysical processes behind the biased MCS precipitation evolutions. For a more conclusive explanation, further in-depth analyses are warranted.

In addition to physical parameterizations, large-scale biases could also contribute to the biased MCS activity. Previous evaluations reported that C192AM4 reliably simulates the synoptic-

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

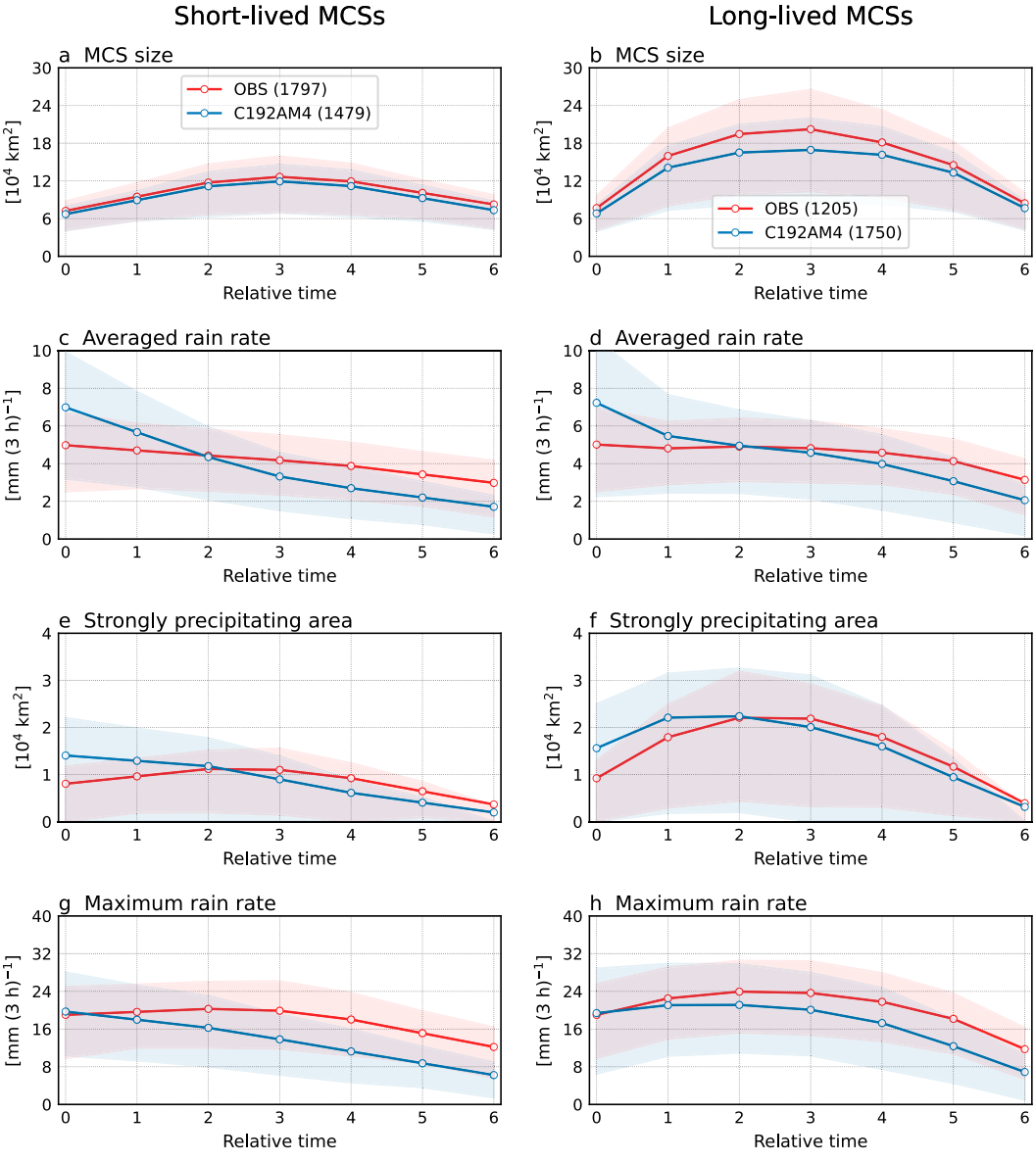

FIG. 13. Composite life cycle evolutions of (a),(b) MCS size, (c),(d) strongly precipitating area, (e),(f) averaged rain rate, and (g),(h) maximum rain rate for (left) short-lived (lifetime $\leq$ 9 h) and (right) long-lived (lifetime $\geq$ 12 h) MCSs in warm season (MAM and JJA). The dry MCSs are excluded. The x axis is the normalized MCS time, and the colored shadings denote the 25th–75th percentile ranges. The MCSs generated in the boxed region in Fig. 1 are considered. Also printed in the legend is the number of the considered MCSs.

to-hemispheric-scale circulations at MCS genesis but exaggerates the North Atlantic subtropical high (Dong et al. 2023; You et al. 2024). You et al. (2024) speculated that this biased North Atlantic subtropical high is attributed to the lack of air–sea interaction in an atmosphere-only model and may be partly responsible for the eastward-shifted MCS distribution. Although not explicitly examined before, misrepresentation of midtropospheric subsynoptic perturbations emanating from the Rocky Mountains (Wang et al. 2009, 2011; Li and Smith 2010; Song et al. 2021) could also contribute to the biased MCS distribution, as also discussed by Dong et al. (2023).

To isolate the large-scale biases, we repeated the analyses using a nudged C192AM4 simulation. In this simulation, the model horizontal winds, temperature, and surface pressure are nudged to the Modern-Era Retrospective Analysis for Research and Applications, version 2 (MERRA-2), reanalysis (Gelaro et al. 2017) with a nudging time scale of 6 h. It is important to note that specific humidity, clouds, and precipitation are still computed interactively. We find that the nudged simulation does not significantly reduce MCS genesis biases compared to the free-simulation result, although the overestimated MCS genesis in the eastern United States is partly

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

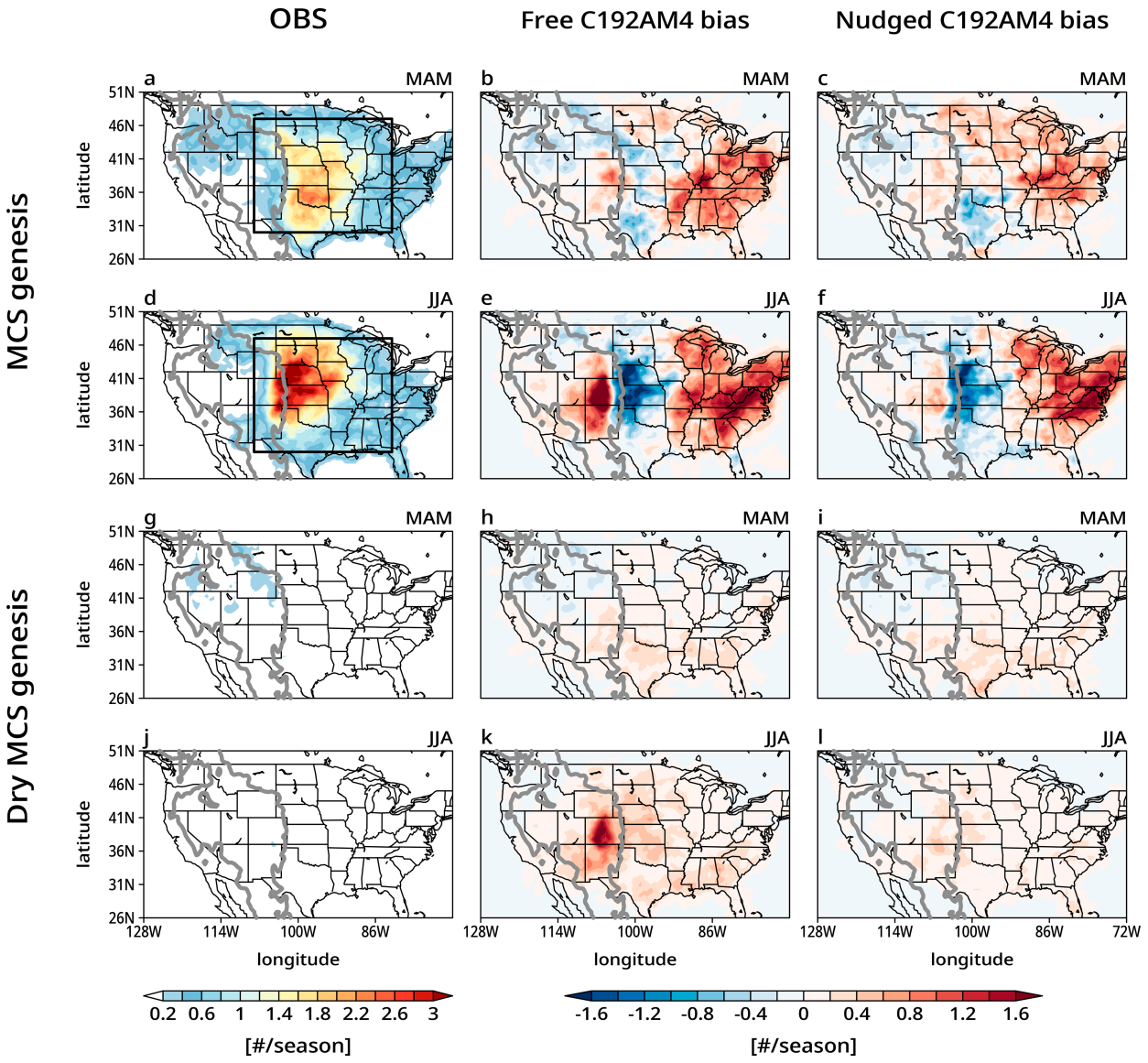

FIG. 14. Genesis frequency of MCSs (number per season) in (a) MAM and (d) JJA, its bias in free C192AM4 simulation in (b) MAM and (e) JJA, and its bias in nudged C192AM4 simulation in (c) MAM and (f) JJA. (g)–(l) As in (a)–(f), but for dry MCSs. The boxed region in (a) and (d) denotes the central United States (identical to that in Fig. 1). The thick gray contour marks the 1200-m terrain height.

resolved (Figs. 14a–f). The MCS precipitation and life cycle characteristics are also not satisfactorily improved (figures not shown). Although reduced from the free simulation, 12.5% and 13.3% of MCSs in spring and summer, respectively, are dry MCSs in the central United States (boxed region in Figs. 14a,d), still larger fractions than observations. One exception is the reduction of summertime spurious MCSs over the eastern Rockies (cf. Figs. 14e,f). Notably, these mountainous MCSs are mostly dry MCSs (cf. Figs. 14k,l). Although not conclusive, it is assumed that a ~50-km horizontal resolution of C192AM4 may still be insufficient to realistically represent the complex terrain of the Rocky Mountains and the adjusted meteorological variables (Wu et al. 2017), causing the misrepresentation of nonprecipitating high clouds. This issue needs to be examined by using higher-resolution AM4 simulations (Lin et al. 2023; Nikumbh et al. 2024).

Last, it is important to acknowledge that there could be some uncertainty in our results in association with the MCS tracking method. On the one hand, MCSs are identified as those with $T_b < 221$ K and area coverage $> 30\,000$ km^2. These thresholds may be too stringent to capture the earliest stages of MCS development, when smaller isolated storms begin to aggregate into a single, deeper convective system. However, this limitation is applied to both observations and model outputs, ensuring at least a fair basis for comparison. On the other hand, given the increasing popularity of the Lagrangian framework in MCS research and the variety of available tracking algorithms with different proxies and thresholds

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

(Huang et al. 2018; Haberlie and Ashley 2019; Cheeks et al. 2020; Prein et al. 2020; Feng et al. 2021a,b; Núñez Ocasio and Moon 2024), it would be valuable to quantify the uncertainty range in MCS statistics arising from the choice of algorithm. A systematic intercomparison of tracking methods, while likely requiring a coordinated community effort [e.g., MCS Tracking Method Intercomparison (MCSMIP); Feng et al. 2025], would support a more robust and comprehensive assessment of MCS characteristics.

Acknowledgments. We appreciate the two anonymous reviewers for their constructive feedback. This study was supported by the U.S. National Oceanic and Atmospheric Administration (NOAA) through Grant NA23OAR4310132. Chanil Park was also supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (RS-2024-00406720). Wenhao Dong was supported by NOAA's Science Collaboration Program and administered by UCAR's Cooperative Programs for the Advancement of Earth System Science (CPAESS) under Awards NA16NWS4620043 and NA18NWS4620043B.

Data availability statement. The CLAUS brightness temperature data are available at https://catalogue.ceda.ac.uk/uuid/c2112bdd5f0ad698e70be6ab54c9a2ac. The MSWEP V2 precipitation data can be accessed at https://www.gloh2o.org/mswep/. The C192AM4 data are obtained from collaborations with Wenhao Dong at GFDL. The source codes of the default GFDL AM4 model can be obtained at https://data1.gfdl.noaa.gov/nomads/forms/am4.0/.

REFERENCES

Ahn, M.-S., D. Kim, S. Park, and Y.-G. Ham, 2019: Do we need to parameterize mesoscale convective organization to mitigate the MJO-mean state trade-off? *Geophys. Res. Lett.*, **46**, 2293–2301, https://doi.org/10.1029/2018GL080314.

Beck, H. E., E. F. Wood, M. Pan, C. K. Fisher, D. G. Miralles, A. I. J. M. van Dijk, T. R. McVicar, and R. F. Adler, 2019: MSWEP V2 global 3-hourly 0.1° precipitation: Methodology and quantitative assessment. *Bull. Amer. Meteor. Soc.*, **100**, 473–500, https://doi.org/10.1175/BAMS-D-17-0138.1.

Cheeks, S. M., S. Fueglistaler, and S. T. Garner, 2020: A satellite-based climatology of central and southeastern U.S. mesoscale convective systems. *Mon. Wea. Rev.*, **148**, 2607–2621, https://doi.org/10.1175/MWR-D-20-0027.1.

Chen, C.-C., J. Richter, C. Liu, M. W. Moncrieff, Q. Tang, W. Lin, S. Xie, and P. J. Rasch, 2021: Effects of organized convection parameterization on the MJO and precipitation in E3SMv1. Part I: Mesoscale heating. *J. Adv. Model. Earth Syst.*, **13**, e2020MS002401, https://doi.org/10.1029/2020MS002401.

Chen, G., and B. P. Kirtman, 2018: Long-lived mesoscale convective systems of superparameterized CAM and the response of CAM. *J. Adv. Model. Earth Syst.*, **10**, 2269–2286, https://doi.org/10.1029/2018MS001339.

Coniglio, M. C., J. Y. Hwang, and D. J. Stensrud, 2010: Environmental factors in the upscale growth and longevity of MCSs derived from rapid update cycle analyses. *Mon. Wea. Rev.*, **138**, 3514–3539, https://doi.org/10.1175/2010MWR3233.1.

Corfidi, S. F., 2003: Cold pools and MCS propagation: Forecasting the motion of downwind-developing MCSs. *Wea. Forecasting*, **18**, 997–1017, https://doi.org/10.1175/1520-0434(2003)018<0997:CPAMPF>2.0.CO;2.

Crook, N. A., and M. W. Moncrieff, 1988: The effect of large-scale convergence on the generation and maintenance of deep moist convection. *J. Atmos. Sci.*, **45**, 3606–3624, https://doi.org/10.1175/1520-0469(1988)045<3606:TEOLSC>2.0.CO;2.

Dai, A., 2006: Precipitation characteristics in eighteen coupled climate models. *J. Climate*, **19**, 4605–4630, https://doi.org/10.1175/JCLI3884.1.

Dong, W., M. Zhao, Y. Ming, and V. Ramaswamy, 2021: Representation of tropical mesoscale convective systems in a general circulation model: Climatology and response to global warming. *J. Climate*, **34**, 5657–5671, https://doi.org/10.1175/JCLI-D-20-0535.1.

——, ——, ——, and ——, 2022: Significant increase in sea surface temperature at the genesis of tropical mesoscale convective systems. *Geophys. Res. Lett.*, **49**, e2022GL101950, https://doi.org/10.1029/2022GL101950.

——, ——, ——, J. P. Krasting, and V. Ramswamy, 2023: Simulation of United States mesoscale convective systems using GFDL's new high-resolution general circulation model. *J. Climate*, **36**, 6967–6990, https://doi.org/10.1175/JCLI-D-22-0529.1.

Doswell, C. A., III, H. E. Brooks, and R. A. Maddox, 1996: Flash flood forecasting: An ingredients-based methodology. *Wea. Forecasting*, **11**, 560–581, https://doi.org/10.1175/1520-0434(1996)011<0560:FFFAIB>2.0.CO;2.

Ellingson, R. G., and R. R. Ferrado, 1983: An examination of a technique for estimating the longwave radiation budget from satellite radiance observations. *J. Climate Appl. Meteor.*, **22**, 1416–1423, https://doi.org/10.1175/1520-0450(1983)022<1416:AEOATF>2.0.CO;2.

Elsaesser, G. S., R. Roca, T. Fiolleau, A. D. Del Genio, and J. Wu, 2022: A simple model for tropical convective cloud shield area growth and decay rates informed by geostationary IR, GPM, and aqua/AIRS satellite data. *J. Geophys. Res. Atmos.*, **127**, e2021JD035599, https://doi.org/10.1029/2021JD035599.

Engerer, N. A., D. J. Stensrud, and M. C. Coniglio, 2008: Surface characteristics of observed cold pools. *Mon. Wea. Rev.*, **136**, 4839–4849, https://doi.org/10.1175/2008MWR2528.1.

Feng, Z., L. R. Leung, R. A. Houze Jr., S. Hagos, J. Hardin, Q. Yang, B. Han, and J. Fan, 2018: Structure and evolution of mesoscale convective systems: Sensitivity to cloud microphysics in convection-permitting simulations over the United States. *J. Adv. Model. Earth Syst.*, **10**, 1470–1494, https://doi.org/10.1029/2018MS001305.

——, R. A. Houze Jr., L. R. Leung, F. Song, J. C. Hardin, J. Wang, W. I. Gustafson Jr., and C. R. Homeyer, 2019: Spatiotemporal characteristics and large-scale environments of mesoscale convective systems east of the Rocky Mountains. *J. Climate*, **32**, 7303–7328, https://doi.org/10.1175/JCLI-D-19-0137.1.

——, and Coauthors, 2021a: A global high-resolution mesoscale convective system database using satellite-derived cloud tops, surface precipitation, and tracking. *J. Geophys. Res. Atmos.*, **126**, e2020JD034202, https://doi.org/10.1029/2020JD034202.

——, F. Song, K. Sakaguchi, and L. R. Leung, 2021b: Evaluation of mesoscale convective systems in climate simulations:

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

Methodological development and results from MPAS-CAM over the United States. *J. Climate*, **34**, 2611–2633, https://doi.org/10.1175/JCLI-D-20-0136.1.

——, L. R. Leung, J. Hardin, C. R. Terai, F. Song, and P. Caldwell, 2023: Mesoscale convective systems in DYAMOND global convection-permitting simulations. *Geophys. Res. Lett.*, **50**, e2022GL102603, https://doi.org/10.1029/2022GL102603.

——, and Coauthors, 2025: Mesoscale Convective Systems Tracking Method Intercomparison (MCSMIP): Application to DYAMOND global km-scale simulations. *J. Geophys. Res. Atmos.*, **130**, e2024JD042204, https://doi.org/10.1029/2024JD042204.

Fritsch, J. M., R. J. Kane, and C. R. Chelius, 1986: The contribution of mesoscale convective weather systems to the warm-season precipitation in the United States. *J. Climate Appl. Meteor.*, **25**, 1333–1345, https://doi.org/10.1175/1520-0450(1986)025<1333:TCOMCW>2.0.CO;2.

Gallus, W. A., Jr., N. A. Snook, and E. V. Johnson, 2008: Spring and summer severe weather reports over the Midwest as a function of convective mode: A preliminary study. *Wea. Forecasting*, **23**, 101–113, https://doi.org/10.1175/2007WAF2006120.1.

Geerts, B., and Coauthors, 2017: The 2015 Plains Elevated Convection at Night field project. *Bull. Amer. Meteor. Soc.*, **98**, 767–786, https://doi.org/10.1175/BAMS-D-15-00257.1.

Gelaro, R., and Coauthors, 2017: The Modern-Era Retrospective Analysis for Research and Applications, version 2 (MERRA-2). *J. Climate*, **30**, 5419–5454, https://doi.org/10.1175/JCLI-D-16-0758.1.

Grabowski, W. W., and P. K. Smolarkiewicz, 1999: CRCP: A cloud resolving convection parameterization for modeling the tropical convecting atmosphere. *Physica D*, **133**, 171–178, https://doi.org/10.1016/S0167-2789(99)00104-9.

Guo, H., Y. Ming, S. Fan, L. Zhou, L. Harris, and M. Zhao, 2021: Two-moment bulk cloud microphysics with prognostic precipitation in GFDL's atmosphere model AM4.0: Configuration and performance. *J. Adv. Model. Earth Syst.*, **13**, e2020MS002453, https://doi.org/10.1029/2020MS002453.

Haarsma, R. J., and Coauthors, 2016: High Resolution Model Intercomparison Project (HighResMIP v1.0) for CMIP6. *Geosci. Model Dev.*, **9**, 4185–4208, https://doi.org/10.5194/gmd-9-4185-2016.

Haberlie, A. M., and W. S. Ashley, 2019: A radar-based climatology of mesoscale convective systems in the United States. *J. Climate*, **32**, 1591–1606, https://doi.org/10.1175/JCLI-D-18-0559.1.

Hartmann, D. L., H. H. Hendon, and R. A. Houze Jr., 1984: Some implications of the mesoscale circulations in tropical cloud clusters for large-scale dynamics and climate. *J. Climate*, **41**, 113–121, https://doi.org/10.1175/1520-0469(1984)041%3C0113:SIOTMC%3E2.0.CO;2.

Hodges, K. I., D. W. Chappell, G. J. Robinson, and G. Yang, 2000: An improved algorithm for generating global window brightness temperatures from multiple satellite infrared imagery. *J. Atmos. Oceanic Technol.*, **17**, 1296–1312, https://doi.org/10.1175/1520-0426(2000)017<1296:AIAFGG>2.0.CO;2.

Houze, R. A., Jr., 1989: Observed structure of mesoscale convective systems and implications for large-scale heating. *Quart. J. Roy. Meteor. Soc.*, **115**, 425–461, https://doi.org/10.1002/qj.49711548702.

——, 1997: Stratiform precipitation in regions of convection: A meteorological paradox? *Bull. Amer. Meteor. Soc.*, **78**, 2179–2196, https://doi.org/10.1175/1520-0477(1997)078<2179:SPIROC>2.0.CO;2.

——, 2004: Mesoscale convective systems. *Rev. Geophys.*, **42**, RG4003, https://doi.org/10.1029/2004RG000150.

——, 2018: 100 years of research on mesoscale convective systems. *A Century of Progress in Atmospheric and Related Sciences: Celebrating the American Meteorological Society Centennial*, *Meteor. Monogr.*, No. **59**, Amer. Meteor. Soc., https://doi.org/10.1175/AMSMONOGRAPHS-D-19-0006.1.

——, S. A. Rutledge, M. I. Biggerstaff, and B. F. Smull, 1989: Interpretation of Doppler weather radar displays of midlatitude mesoscale convective systems. *Bull. Amer. Meteor. Soc.*, **70**, 608–619, https://doi.org/10.1175/1520-0477(1989)070<0608:IODWRD>2.0.CO;2.

Hsu, W.-C., G. J. Kooperman, W. M. Hannah, K. A. Reed, A. A. Akinsanola, and A. G. Pendergrass, 2023: Evaluating mesoscale convective systems over the US in conventional and multiscale modeling framework configurations of E3SMv1. *J. Geophys. Res. Atmos.*, **128**, e2023JD038740, https://doi.org/10.1029/2023JD038740.

Hu, H., Z. Feng, and L.-Y. R. Leung, 2021: Linking flood frequency with mesoscale convective systems in the US. *Geophys. Res. Lett.*, **48**, e2021GL092546, https://doi.org/10.1029/2021GL092546.

Huang, X., C. Hu, X. Huang, Y. Chu, Y-h Tseng, G. J. Zhang, and Y. Lin, 2018: A long-term tropical mesoscale convective systems dataset based on a novel objective automatic tracking algorithm. *Climate Dyn.*, **51**, 3145–3159, https://doi.org/10.1007/s00382-018-4071-0.

Jakob, C., and S. A. Klein, 2000: A parameterization of the effects of cloud and precipitation overlap for use in general-circulation models. *Quart. J. Roy. Meteor. Soc.*, **126**, 2525–2544, https://doi.org/10.1002/qj.49712656809.

Janowiak, J. E., R. J. Joyce, and Y. Yarosh, 2001: A real–time global half-hourly pixel-resolution infrared dataset and its applications. *Bull. Amer. Meteor. Soc.*, **82**, 205–217, https://doi.org/10.1175/1520-0477(2001)082<0205:ARTGHH>2.3.CO;2.

Jirak, I. L., W. R. Cotton, and R. L. McAnelly, 2003: Satellite and radar survey of mesoscale convective system development. *Mon. Wea. Rev.*, **131**, 2428–2449, https://doi.org/10.1175/1520-0493(2003)131<2428:SARSOM>2.0.CO;2.

Johns, R. H., and C. A. Doswell III, 1992: Severe local storms forecasting. *Wea. Forecasting*, **7**, 588–612, https://doi.org/10.1175/1520-0434(1992)007<0588:SLSF>2.0.CO;2.

Kim, D., A. H. Sobel, A. D. Del Genio, Y. Chen, S. J. Camargo, M.-S. Yao, M. Kelley, and L. Nazarenko, 2012: The tropical subseasonal variability simulated in the NASA GISS general circulation model. *J. Climate*, **25**, 4641–4659, https://doi.org/10.1175/JCLI-D-11-00447.1.

Klein, S. A., X. Jiang, J. Boyle, S. Malyshev, and S. Xie, 2006: Diagnosis of the summertime warm and dry bias over the U.S. Southern Great Plains in the GFDL climate model using a weather forecasting approach. *Geophys. Res. Lett.*, **33**, L18805, https://doi.org/10.1029/2006GL027567.

Konstali, K., C. Spensberger, T. Spengler, and A. Sorteberg, 2024: Global attribution of precipitation to weather features. *J. Climate*, **37**, 1181–1196, https://doi.org/10.1175/JCLI-D-23-0293.1.

Leung, L. R., and Coauthors, 2022: Exploratory precipitation metrics: Spatiotemporal characteristics, process-oriented, and phenomena-based. *J. Climate*, **35**, 3659–3686, https://doi.org/10.1175/JCLI-D-21-0590.1.

Li, L., W. Li, and Y. Kushnir, 2012: Variation of the North Atlantic subtropical high western ridge and its implication to southeastern US summer precipitation. *Climate Dyn.*, **39**, 1401–1412, https://doi.org/10.1007/s00382-011-1214-y.

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

Li, Y., and R. B. Smith, 2010: The detection and significance of diurnal pressure and potential vorticity anomalies east of the Rockies. *J. Atmos. Sci.*, **67**, 2734–2751, https://doi.org/10.1175/2010JAS3423.1.

Lin, G., J. Fan, Z. Feng, W. I. Gustafson Jr., P.-L. Ma, and K. Zhang, 2019: Can the Multiscale Modeling Framework (MMF) simulate the MCS-associated precipitation over the central United States? *J. Adv. Model. Earth Syst.*, **11**, 4669–4686, https://doi.org/10.1029/2019MS001849.

——, C. R. Jones, L. R. Leung, Z. Feng, and M. Ovchinnikov, 2022: Mesoscale convective systems in a superparameterized E3SM simulation at high resolution. *J. Adv. Model. Earth Syst.*, **14**, e2021MS002660, https://doi.org/10.1029/2021MS002660.

Lin, P., Y. Ming, and T. Robinson, 2023: On the resolution sensitivity of equatorial precipitation in a GFDL global atmospheric model. *J. Adv. Model. Earth Syst.*, **15**, e2022MS003300, https://doi.org/10.1029/2022MS003300.

Lin, Y., W. Dong, M. Zhang, Y. Xie, W. Xue, J. Huang, and Y. Luo, 2017: Causes of model dry and warm bias over central U.S. and impact on climate projections. *Nat. Commun.*, **8**, 881, https://doi.org/10.1038/s41467-017-01040-2.

Liu, C., and Coauthors, 2017: Continental-scale convection-permitting modeling of the current and future climate of North America. *Climate Dyn.*, **49**, 71–95, https://doi.org/10.1007/s00382-016-3327-9.

Loehrer, S. M., and R. H. Johnson, 1995: Surface pressure and precipitation life cycle characteristics of pre-STORM mesoscale convective systems. *Mon. Wea. Rev.*, **123**, 600–621, https://doi.org/10.1175/1520-0493(1995)123<0600:SPAPLC>2.0.CO;2.

Machado, L. A. T., W. B. Rossow, R. L. Guedes, and A. W. Walker, 1998: Life cycle variations of mesoscale convective systems over the Americas. *Mon. Wea. Rev.*, **126**, 1630–1654, https://doi.org/10.1175/1520-0493(1998)126<1630:LCVOMC>2.0.CO;2.

Maddox, R. A., 1980: Meoscale convective complexes. *Bull. Amer. Meteor. Soc.*, **61**, 1374–1400, https://doi.org/10.1175/1520-0477(1980)061<1374:MCC>2.0.CO;2.

Moncrieff, M. W., 2019: Toward a dynamical foundation for organized convection parameterization in GCMs. *Geophys. Res. Lett.*, **46**, 14 103–14 108, https://doi.org/10.1029/2019GL085316.

——, C. Liu, and P. Bogenschutz, 2017: Simulation, modeling, and dynamically based parameterization of organized tropical convection for global climate models. *J. Atmos. Sci.*, **74**, 1363–1380, https://doi.org/10.1175/JAS-D-16-0166.1.

Moore, B. J., P. J. Neiman, F. M. Ralph, and F. E. Barthold, 2012: Physical processes associated with heavy flooding rainfall in Nashville, Tennessee, and vicinity during 1–2 May 2010: The role of an atmospheric river and mesoscale convective systems. *Mon. Wea. Rev.*, **140**, 358–378, https://doi.org/10.1175/MWR-D-11-00126.1.

Murakami, H., T. L. Delworth, W. F. Cooke, M. Zhao, B. Xiang, and P.-C. Hsu, 2020: Detected climatic change in global distribution of tropical cyclones. *Proc. Natl. Acad. Sci. USA*, **117**, 10 706–10 714, https://doi.org/10.1073/pnas.1922500117.

Na, Y., Q. Fu, L. R. Leung, C. Kodama, and R. Lu, 2022: Mesoscale convective systems simulated by a high-resolution global nonhydrostatic model over the United States and China. *J. Geophys. Res. Atmos.*, **127**, e2021JD035916, https://doi.org/10.1029/2021JD035916.

Neelin, J. D., and Coauthors, 2023: Process-oriented diagnostics: Principles, practice, community development, and common standards. *Bull. Amer. Meteor. Soc.*, **104**, E1452–E1468, https://doi.org/10.1175/BAMS-D-21-0268.1.

Nesbitt, S. W., R. Cifelli, and S. A. Rutledge, 2006: Storm morphology and rainfall characteristics of TRMM precipitation features. *Mon. Wea. Rev.*, **134**, 2702–2721, https://doi.org/10.1175/MWR3200.1.

Nikumbh, A. C., P. Lin, D. Paynter, and Y. Ming, 2024: Does increasing horizontal resolution improve the simulation of intense tropical rainfall in GFDL's AM4 model? *Geophys. Res. Lett.*, **51**, e2023GL106708, https://doi.org/10.1029/2023GL106708.

Norman, M. R., and Coauthors, 2021: Unprecedented cloud resolution in a GPU-enabled full-physics atmospheric climate simulation on OLCF's summit supercomputer. *Int. J. High Perform. Comput. Appl.*, **36**, 93–105, https://doi.org/10.1177/10943420211027539.

Núñez Ocasio, K. M., and Z. L. Moon, 2024: TAMS: A tracking, classifying, and variable-assigning algorithm for mesoscale convective systems in simulated and satellite-derived datasets. *Geosci. Model Dev.*, **17**, 6035–6049, https://doi.org/10.5194/gmd-17-6035-2024.

Ohring, G., A. Gruber, and R. Ellingson, 1984: Satellite determinations of the relationship between total longwave radiation flux and infrared window radiance. *J. Climate Appl. Meteor.*, **23**, 416–425, https://doi.org/10.1175/1520-0450(1984)023<0416:SDOTRB>2.0.CO;2.

Park, C., Yi. Ming, Yi. Deng, and W. Dong, 2025: Process-based quantification of synoptic forced uplift for springtime MCS initiation over the U.S. Great Plains. *J. Geophys. Res. Atmos.*, **130**, e2025JD044460,https://doi.org/10.1029/2025JD044460.

Park, S., C. Song, S. Kim, and J. Kim, 2024: Parameterization of the elevated convection with a unified convection scheme (UNICON) and its impacts on the diurnal cycle of precipitation. *J. Adv. Model. Earth Syst.*, **16**, e2023MS003651, https://doi.org/10.1029/2023MS003651.

Peters, J. M., and R. S. Schumacher, 2014: Objective categorization of heavy-rain-producing MCS synoptic types by rotated principal component analysis. *Mon. Wea. Rev.*, **142**, 1716–1737, https://doi.org/10.1175/MWR-D-13-00295.1.

Prein, A. F., and Coauthors, 2015: A review on regional convection-permitting climate modeling: Demonstrations, prospects, and challenges. *Rev. Geophys.*, **53**, 323–361, https://doi.org/10.1002/2014RG000475.

——, C. Liu, K. Ikeda, R. Bullock, R. M. Rasmussen, G. J. Holland, and M. Clark, 2020: Simulating North American mesoscale convective systems with a convection-permitting climate model. *Climate Dyn.*, **55**, 95–110, https://doi.org/10.1007/s00382-017-3993-2.

——, and Coauthors, 2024: Km-scale simulations of mesoscale convective systems over South America—A feature tracker intercomparison. *J. Geophys. Res. Atmos.*, **129**, e2023JD040254 and, https://doi.org/10.1029/2023JD040254.

Rasp, S., T. Selz, and G. C. Craig, 2016: Convective and slantwise trajectory ascent in convection-permitting simulations of midlatitude cyclones. *Mon. Wea. Rev.*, **144**, 3961–3976, https://doi.org/10.1175/MWR-D-16-0112.1.

Raymond, D. J., and H. Jiang, 1990: A theory for long-lived mesoscale convective systems. *J. Atmos. Sci.*, **47**, 3067–3077, https://doi.org/10.1175/1520-0469(1990)047<3067:ATFLLM>2.0.CO;2.

Roberts, M. J., and Coauthors, 2018: The benefits of global high resolution for climate simulation: Process understanding and the enabling of stakeholder decisions at the regional scale.

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

Bull. Amer. Meteor. Soc., **99**, 2341–2359, https://doi.org/10.1175/BAMS-D-15-00320.1.

Roca, R., J. Aublanc, P. Chambon, T. Fiolleau, and N. Viltard, 2014: Robust observational quantification of the contribution of mesoscale convective systems to rainfall in the tropics. *J. Climate*, **27**, 4952–4958, https://doi.org/10.1175/JCLI-D-13-00628.1.

Rodgers, D. M., M. J. Magnano, and J. H. Arns, 1985: Mesoscale convective complexes over the United States during 1983. *Mon. Wea. Rev.*, **113**, 888–901, https://doi.org/10.1175/1520-0493(1985)113<0888:MCCOTU>2.0.CO;2.

Rotstayn, L. D., 1997: A physically based scheme for the treatment of stratiform clouds and precipitation in large-scale models. I: Description and evaluation of the microphysical processes. *Quart. J. Roy. Meteor. Soc.*, **123**, 1227–1282, https://doi.org/10.1002/qj.49712354106.

Rotunno, R., J. B. Klemp, and M. L. Weisman, 1988: A theory for strong, long-lived squall lines. *J. Atmos. Sci.*, **45**, 463–485, https://doi.org/10.1175/1520-0469(1988)045<0463:ATFSLL>2.0.CO;2.

Schaller, N., I. Mahlstein, J. Cermak, and R. Knutti, 2011: Analyzing precipitation projections: A comparison of different approaches to climate model evaluation. *J. Geophys. Res.*, **116**, D10118, https://doi.org/10.1029/2010JD014963.

Schumacher, C., R. A. Houze Jr., and I. Kraucunas, 2004: The tropical dynamical response to latent heating estimates derived from the TRMM precipitation radar. *J. Atmos. Sci.*, **61**, 1341–1358, https://doi.org/10.1175/1520-0469(2004)061%3C1341:TTDRTL%3E2.0.CO;2.

Schumacher, R. S., and R. H. Johnson, 2006: Characteristics of U.S. extreme rain events during 1999–2003. *Wea. Forecasting*, **21**, 69–85, https://doi.org/10.1175/WAF900.1.

——, and K. L. Rasmussen, 2020: The formation, character and changing nature of mesoscale convective systems. *Nat. Rev. Earth Environ.*, **1**, 300–314, https://doi.org/10.1038/s43017-020-0057-7.

Smull, B. F., and R. A. Houze Jr., 1987: Rear inflow in squall lines with trailing stratiform precipitation. *Mon. Wea. Rev.*, **115**, 2869–2889, https://doi.org/10.1175/1520-0493(1987)115<2869:RIISLW>2.0.CO;2.

Song, F., Z. Feng, L. R. Leung, R. A. Houze Jr., J. Wang, J. Hardin, and C. R. Homeyer, 2019: Contrasting spring and summer large-scale environments associated with mesoscale convective systems over the U.S. Great Plains. *J. Climate*, **32**, 6749–6767, https://doi.org/10.1175/JCLI-D-18-0839.1.

——, ——, ——, B. Pokharel, S.-Y. Wang, X. Chen, K. Sakaguchi, and C.-C. Wang, 2021: Crucial roles of eastward propagating environments in the summer MCS initiation over the U.S. Great Plains. *J. Geophys. Res. Atmos.*, **126**, e2021JD034991, https://doi.org/10.1029/2021JD034991.

Stensrud, D. J., 1996: Effects of persistent, midlatitude mesoscale regions of convection on the large-scale environment during the warm season. *J. Atmos. Sci.*, **53**, 3503–3527, https://doi.org/10.1175/1520-0469(1996)053<3503:EOPMMR>2.0.CO;2.

——, and J. L. Anderson, 2001: Is midlatitude convection an active or a passive player in producing global circulation patterns? *J. Climate*, **14**, 2222–2237, https://doi.org/10.1175/1520-0442(2001)014<2222:IMCAAO>2.0.CO;2.

Swenson, L. M., 2023: Can simple metrics identify the process(es) driving extreme precipitation? *Climate*, **11**, 88, https://doi.org/10.3390/cli11040088.

Tang, S., P. Gleckler, S. Xie, J. Lee, M.-S. Ahn, C. Covey, and C. Zhang, 2021: Evaluating the diurnal and semidiurnal cycle of precipitation in CMIP6 models using satellite- and ground-based observations. *J. Climate*, **34**, 3189–3210, https://doi.org/10.1175/JCLI-D-20-0639.1.

Tao, W.-K., and M. Moncrieff, 2009: Multiscale cloud system modeling. *Rev. Geophys.*, **47**, RG4002, https://doi.org/10.1029/2008RG000276.

Tapiador, F. J., R. Roca, A. Del Genio, B. Dewitte, W. Petersen, and F. Zhang, 2019: Is precipitation a good metric for model performance? *Bull. Amer. Meteor. Soc.*, **100**, 223–233, https://doi.org/10.1175/BAMS-D-17-0218.1.

Trenberth, K. E., A. Dai, R. M. Rasmussen, and D. B. Parsons, 2003: The changing character of precipitation. *Bull. Amer. Meteor. Soc.*, **84**, 1205–1218, https://doi.org/10.1175/BAMS-84-9-1205.

——, L. Smith, T. Qian, A. Dai, and J. Fasullo, 2007: Estimates of the global water budget and its annual cycle using observational and model data. *J. Hydrometeor.*, **8**, 758–769, https://doi.org/10.1175/JHM600.1.

Tsai, W.-L., and Coauthors, 2025: Co-occurring atmospheric features and their contributions to precipitation extremes. *J. Geophys. Res. Atmos.*, **130**, e2024JD041687, https://doi.org/10.1029/2024JD041687.

Wang, S.-Y., T.-C. Chen, and S. E. Taylor, 2009: Evaluations of NAM forecasts on midtropospheric perturbation-induced convective storms over the U.S. northern plains. *Wea. Forecasting*, **24**, 1309–1333, https://doi.org/10.1175/2009WAF2222185.1.

——, ——, and E. S. Takle, 2011: Climatology of summer midtropospheric perturbations in the US Northern plains. Part II: Large-scale effects of the Rocky Mountains on genesis. *Climate Dyn.*, **36**, 1221–1237, https://doi.org/10.1007/s00382-010-0765-7.

Weisman, M. L., J. B. Klemp, and R. Rotunno, 1988: Structure and evolution of numerically simulated squall lines. *J. Atmos. Sci.*, **45**, 1990–2013, https://doi.org/10.1175/1520-0469(1988)045<1990:SAEONS>2.0.CO;2.

Wilson, J. W., and R. D. Roberts, 2006: Summary of convective storm initiation and evolution during IHOP: Observational and modeling perspective. *Mon. Wea. Rev.*, **134**, 23–47, https://doi.org/10.1175/MWR3069.1.

Wu, C., X. Liu, Z. Lin, A. M. Rhoades, P. A. Ullrich, C. M. Zarzycki, Z. Lu, and S. R. Rahimi-Esfarjani, 2017: Exploring a variable-resolution approach for simulating regional climate in the Rocky Mountain region using the VR-CESM. *J. Geophys. Res. Atmos.*, **122**, 10 939–10 965, https://doi.org/10.1002/2017JD027008.

Yang, Q., R. A. Houze Jr., L. R. Leung, and Z. Feng, 2017: Environments of long-lived mesoscale convective systems over the central United States in convection permitting climate simulations. *J. Geophys. Res. Atmos.*, **122**, 13 288–13 307, https://doi.org/10.1002/2017JD027033.

——, A. J. Majda, and M. W. Moncrieff, 2019: Upscale impact of mesoscale convective systems and its parameterization in an idealized GCM for an MJO analog above the equator. *J. Atmos. Sci.*, **76**, 865–892, https://doi.org/10.1175/JAS-D-18-0260.1.

You, Z., Y. Deng, Y. Ming, and W. Dong, 2024: A multiscale assessment of the springtime U.S. mesoscale convective systems in the NOAA GFDL AM4. *Climate Dyn.*, **62**, 4017–4030, https://doi.org/10.1007/s00382-024-07114-4.

Zhang, M., and Coauthors, 2024: Mesoscale convective systems represented in high resolution E3SMv2 and impact of new cloud and convection parameterizations. *J. Geophys. Res.*

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC

Atmos., **129**, e2024JD040828, https://doi.org/10.1029/2024JD040828.

Zhang, Z., A. Varble, Z. Feng, J. C. Hardin, and E. Zipser, 2021: Growth of mesoscale convective systems in observations and a seasonal convection-permitting simulation over Argentina. *Mon. Wea. Rev.*, **149**, 3469–3490, https://doi.org/10.1175/MWR-D-20-0411.1.

Zhao, M., 2020: Simulations of atmospheric rivers, their variability, and response to global warming using GFDL's new high-resolution general circulation model. *J. Climate*, **33**, 10 287–10 303, https://doi.org/10.1175/JCLI-D-20-0241.1.

——, 2022: A study of AR-, TS-, and MCS-associated precipitation and extreme precipitation in present and warmer climates. *J. Climate*, **35**, 479–497, https://doi.org/10.1175/JCLI-D-21-0145.1.

——, 2024: Cloud radiative effects associated with daily weather regimes. *Geophys. Res. Lett.*, **51**, e2024GL109090, https://doi.org/10.1029/2024GL109090.

——, I. M. Held, S.-J. Lin, and G. A. Vecchi, 2009: Simulations of global hurricane climatology, interannual variability, and response to global warming using a 50-km resolution GCM. *J. Climate*, **22**, 6653–6678, https://doi.org/10.1175/2009JCLI3049.1.

——, ——, and G. A. Vecchi, 2010: Retrospective forecasts of the hurricane season using a global atmospheric model assuming persistence of SST anomalies. *Mon. Wea. Rev.*, **138**, 3858–3868, https://doi.org/10.1175/2010MWR3366.1.

——, and Coauthors, 2018a: The GFDL global atmosphere and land model AM4.0/LM4.0: 1. Simulation characteristics with prescribed SSTs. *J. Adv. Model. Earth Syst.*, **10**, 691–734, https://doi.org/10.1002/2017MS001208.

——, and Coauthors, 2018b: The GFDL global atmosphere and land model AM4.0/LM4.0: 2. Model description, sensitivity studies, and tuning strategies. *J. Adv. Model. Earth Syst.*, **10**, 735–769, https://doi.org/10.1002/2017MS001209.

Zorzetto, E., and L. Li, 2021: Impacts of the North Atlantic subtropical high on daily summer precipitation over the conterminous United States. *J. Hydrometeor.*, **22**, 1697–1712, https://doi.org/10.1175/JHM-D-20-0242.1.

Unauthenticated | Downloaded 11/02/26 07:14 PM UTC